
Parameter-Distribution Graphs as Smooth Manifolds of Neural Architecture Realizable Distributions with NTK-induced Riemannian Metric

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Abstract

Recent advances in drifting models by Deng et al. [2026] assert that distribution evolution can be reframed in terms of iterative neural network training insofar as the function defined by the network can be viewed as a pushforward functional from a prior distribution to another distribution. This paper proves that, under a set of assumptions, the space of distributions that a neural architecture can reach by varying its parameters forms locally immersed manifolds of finite dimensions whose union is an **architecture-realizable distribution locus** of that neural architecture. We carefully demonstrate this conclusion by integrating topics such as differential geometry and Otto calculus. Using concepts from Onsager geometry, we then discuss that the neural tangent kernel of that architecture induces a Riemannian metric on the **parameter–distribution graph**, a genuine, globally embedded submanifold of the product between the parameter space and the ambient probability-density manifold that encodes both the distributions and the parameter information that they correspond to. Then, we show that there exists an immersion relation between the manifolds of different neural architectures, which is an application of this geometric framework. We then performed an empirical experiment to verify our claim that the NTK-induced metric is equivalent to the Euclidean metric on the parameter space, indicating that the metric does capture the architectural effort to realize that evolution. This paper ultimately shows neural architecture-guided distribution evolution is now characterized by differential geometry, a conclusion that responds to the call for future work in the final paragraph of Deng et al. [2026].

Mathematics is the art of giving the same names to different things. — Henri Poincaré, French geometer

Notations and Assumptions

This paper will hereon use the following notations. For a feedforward neural network, $\mathbb{R}^m$ denotes the latent or input space of the neural network, $\mathbb{R}^p$ denotes the parameter space, $\mathbb{R}^d$ the output space. The set of parameters specific to that network is denoted as $\theta \in \mathbb{R}^p$. Thus, m , p , and d would, respectively, be the dimension for the input space, the number of parameters of that neural network, and the dimension of output space. A sample from the input space is denoted as $\epsilon \in \mathbb{R}^m$. We view the neural network parametrized by a parameter as a function denoted by $f_\theta : \mathbb{R}^m \rightarrow \mathbb{R}^d$. q_θ denotes the probability distribution obtained by pushing forward θ , as will be discussed later. For this paper, we make the assumption that the neural architecture used is smooth with respect to its gradients, and all activation functions used are smooth. We further assume that its parameter Jacobian is uniformly square-integrable.² For purposes discussed later, while Deng et al. [2026]’s paper asserts that $m = d$ need not be true, we require $m \geq d$ and strongly prefer each hidden layer having roughly equal widths to prevent rank-restricting bottlenecks.

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²See appendix 9.1 for a definition of and more discussion on this.

1 Introduction

The final paragraph of Deng et al. [2026]’s paper hints at a very promising direction, one that is conceptually inspiring: distribution evolution can be viewed not only in terms of the simulation of differential equations but also in terms of neural network training. This line of thought naturally would inspire the question as to how exactly the two can be integrated under one coherent conceptual picture. To restate Deng et al.’s principal idea: since neural network training via gradient descent is inherently iterative, every step of training induces a functional that "pushes" a designated prior distribution towards the final distribution by sending samples from that prior through that set of induced functions:

$$q_t = [f_t]_{\#} p_{\text{prior}}, \forall t \in [T] \quad (1)$$

Reframed as such, the task of flow matching and diffusion models, and arguably of generative modeling as a whole, is then reduced to simply finding the appropriate pushforward functional f such that $f_{\#} p_{\text{prior}} \approx p_{\text{data}}$, the desired complex distribution we wish to learn. This picture of viewing distribution evolution completely alters simulation of ODE’s / PDE’s at inference time into a one-step inference, reallocating all the distribution transport to training time.

To capitalize on this novel way of viewing distribution evolution, so that it can realize distributions that are actually desired, Deng et al. then introduce the concept of **drift fields**. Drift fields are conceptually the residual of neural network training: the intended difference between the network output at the next iteration and its output at the previous step under the same input sample. The drift field can be thought of as a kernel-based function, $\mathbf{V}_{p,q} : \mathbb{R}^d \rightarrow \mathbb{R}^d$, that pushes the output of a neural network at one step of iteration towards the direction of the final, desired distribution:

$$f_{\theta_{t+1}}(\epsilon) = f_{\theta_t}(\epsilon) + \mathbf{V}_{p,q_{\theta_t}}(f_{\theta_t}(\epsilon)) \quad (2)$$

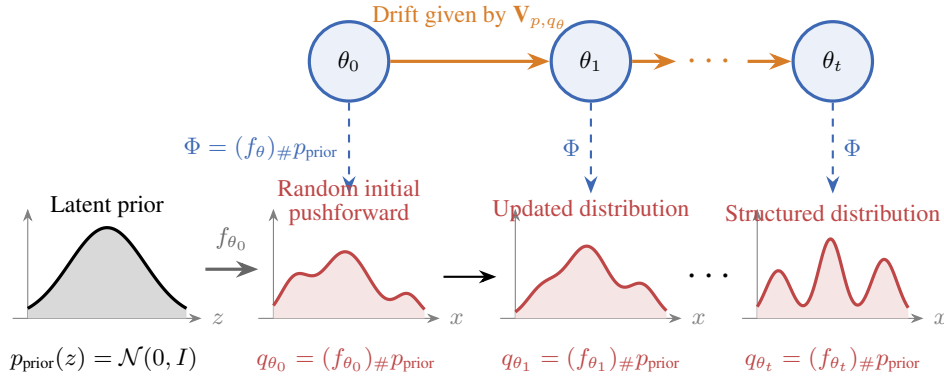


Figure 1: A fixed Gaussian latent prior $p = \mathcal{N}(0, I)$ is pushed forward through the neural generator. Random initialization produces the generally non-Gaussian distribution $q_{\theta_0} = (f_{\theta_0})_{\#} p$. Drift-driven parameter updates then generate the sequence $q_{\theta_0}, q_{\theta_1}, \dots, q_{\theta_t}$.

Neural network evolution, however, cannot perfectly realize every drift field. To quantitatively illustrate this, let us first define the velocity field of a neural network as follows. For a neural network whose parameters are smoothly parameterized by $\theta(t) : [0, T] \rightarrow \mathbb{R}^p$, we define

$$u_t(f_{\theta(t)}(\epsilon)) = \frac{d}{dt} f_{\theta(t)}(\epsilon) \quad (3)$$

With this quantity of velocity drift defined, we can reframe our earlier question as

$$u_t(f_{\theta(t)}(\epsilon)) \stackrel{?}{=} \mathbf{V}_{p,q_t}(f_{\theta(t)}(\epsilon)) \quad (4)$$

If this were true, then the velocity field induced by neural network training guided by a drift field would coincide exactly with the desired drift field on the output space. In that case, the neural network would realize the prescribed transport direction without any architectural distortion. But, as discussed, this simple and clean hypothesized relationship is certainly almost always not true, since a neural network cannot perfectly, in general, exactly drift on the output space in the direction the drift field instructs it to drift. Thus, the natural question would then become: given the tight directional constraints of a neural network, how can the effect of its training be related to a drift field, external to that geometry, that is guiding it? What characterizes the geometric and topological properties of distributional motion of neural architectures when guided by a drift field? This paper seeks to first answer this question, and, along the way, discuss the implications of our answer based on neural tangent kernel geometry. We believe our work establishes a general geometric framework for neural architecture based distribution evolution as a whole. To this end, we will first discuss related work in this field and then introduce some preliminary mathematical tools. Then, we carefully derive and arrive at our central thesis that both sides in 4 are related by the neural tangent kernel specific to that neural network. We will then discuss an important application of this finding: the neural tangent kernel thereby induces a metric on the distribution-level geometry of the realizable distributions, a conclusion that sheds important insight on the hint given in the final paragraph of Deng et al. [2026]’s work.

2 Related Work and Preliminary Concepts

Flow Matching and Distribution Evolution

At the core, the framework our paper proposes is revealed by the principle of flow matching, a central problem in generative modeling concerned with transforming samples from one distribution into samples from the final intended distribution Lipman et al. [2023], or, more generally, the transformation of an entire distribution into another. Distribution evolution, of which flow matching is a method, is not only one specific paradigm of generative modeling, but also arguably the very essence and generalized way of viewing generative models as a whole, since the ultimate task of generative models is the representation of a complex probability distribution through different ways of transformation, at either training or inference time.

Our paper does not seek to study or advance flow matching itself, e.g. proposing a new, specific algorithm for flow matching. Rather, we intend to show the underlying geometric properties of flow matching, thereby revealing what final distributions are ultimately realizable by the geometry of a given neural architecture. So, all in all, flow matching is the central motivating concept of our work as a whole, insofar as conceptually what we do here generalizes it within a broader architecture-induced geometric framework.

Generative Model via Drifting

Flow matching, however, as a principle alone does not require what we do here. The generative model via drifting that Deng et al. [2026] proposes, rather, is the necessary conceptual bridge between flow matching in general and the neural architecture geometry we propose here. Once distribution evolution is reframed as neural architecture drifting, it will necessarily require an understanding of neural architecture dependent geometry.

At its essence, generative model via drifting is a method of realizing flow matching through a one-step simulation supported by multi-step training by capitalizing upon the iterative nature of the training of neural architectures. Generative model via drifting works by defining a drifting field that is based upon the mean-shift method from statistics to "push" samples away from the distribution it is currently at and towards the final, desired distribution.

Generative model trained via drifting in this manner will then inevitably depend upon the neural-architecture dependent geometry, since the neural architecture now is the underlying mechanism that constrains the direction of distributional change at every iterative step of training. The directional and geometric constraints of the architecture, thus, are factors that cannot be ignored if we wish to study the geometric properties of the distributions it induces. This is the central insight that motivates our work.

Wasserstein Gradient Flow

Another crucial concept that would be illuminating to our framework would be Wasserstein gradient flow, a mathematical framework for the description of transportation of probability distributions on a space driven by some notion of transport cost Mokrov et al. [2023]. There, in fact, is existing work that relates Wasserstein gradient flow with the drifting model paradigm Gretton et al. [2026]. This cost is typically defined in terms of the Wasserstein distance between the initial and intended distributions. Wasserstein gradient flow takes advantage of the **continuity equation** from transport theory

$$\partial_t q_t + \nabla \cdot (q_t u_t) = 0, \quad (5)$$

which is the partial differential equation that describes the most general notion of transport for any conserved quantity. This is an equation that will appear at various times in our theory.

Conceptually, the crucial move that Wasserstein gradient flow makes is the redefining of the velocity field in terms of the gradient of some discrepancy function that measures the statistical divergence between the current distribution q_t and the final distribution. In the language of variational calculus, it is defined as

$$u_t = -\nabla_x \frac{\delta F}{\delta q_t}. \quad (6)$$

The relevance of Wasserstein gradient flow to our work is twofold. First, it provides the geometric language through which probability distributions may be treated as points on a space equipped with a transport-induced geometry. Second, it clarifies the way in which our framework departs from the classical gradient-flow setting. In our formulation, the drift field need not arise as the gradient of a scalar functional; instead, a general drift field is filtered through the architecture-dependent NTK operator. In this sense, our framework can be viewed as extending the Wasserstein-gradient-flow perspective toward a more general architecture-dependent transport geometry.

Differential Geometry and Riemannian Manifolds

Differential geometry provides the main mathematical language for characterizing both the distributions a neural network is able to reach and the distributional changes induced by drift-field-guided iterative training of that network. We also use topological notions such as openness, isometries, and density to characterize several relevant spaces. Manifold theory is suited to this purpose because it permits us to study local directions of change and paths through a space of distributions. When a suitable Riemannian metric is available, it additionally provides quantified notions of length and angle, as well as further geometric structure. Although differential geometry is entirely independent of machine learning and deep learning, applying its constructions to architecture-realizable distributions can clarify which changes are locally available to an architecture and how its training trajectories move through distribution space and also enable quantitative characterizations of those trajectories.

A smooth manifold is a topological space equipped with a smooth atlas, (U_i, ϕ_i) , where $\phi_i : U_i \rightarrow \mathbb{R}^D$: a collection of compatible charts mapping open neighborhoods to open subsets of a Euclidean space, or of a Hilbert space in the Hilbert manifold setting. A Riemannian manifold, (M, g) is a smooth manifold equipped with a Riemannian metric, namely a smoothly varying inner product on its tangent spaces, $g(p) : T_p M \times T_p M \rightarrow \mathbb{R}$. A Riemannian manifold would be an apt choice of mathematical object for our purpose here since it is both sufficiently general such that architecture generated distributions can be conformed therein but also sufficiently specific so to as enable accurate geometric characterizations of the behavior of those distributions. On a manifold of distributions, a point represents a distribution, a curve represents a path of distributions, and a tangent vector represents an infinitesimal change at one distribution. This paper will develop the connection between differential geometry and deep learning gradually, and it is this task that anchors this paper. All in all, differential geometry is the principal domain in mathematics we employ for our purpose.

Otto Calculus

However, Riemannian manifolds themselves generally do not have the interpretation of being spaces of probability distributions. A point on a Riemannian manifold may admit different interpretations depending

on the context in which that manifold is defined, but a point being interpreted as an entire distribution is uncommon. So, if we employ differential geometry to study the space of distributions, distributional space must fit within the framework of differentiable manifolds. Otto calculus provides a differential geometric framework for studying the space of probability measures Otto [2001]. Thus, Otto calculus as a framework is indispensable for our work, since our exposition entirely relies on the assumption that the space of distributions on the output space of a neural architecture forms an ambient space from which submanifolds can be constructed. In our paper, we further adopt Lott [2007]’s work on the various geometric calculations on the Wasserstein space, which prove essential to our geometric construction. Let

$$\mathcal{P}_2(\mathbb{R}^d) = \{\mu \in \mathcal{P}(\mathbb{R}^d) : \int_X \|x\|^2 d\mu(x) < \infty\}, \text{ where} \quad (7)$$

$\mathcal{P}_2(\mathbb{R}^d)$ denote the space of Borel probability measures on $\mathbb{R}^d$ with finite second moment, endowed with the 2-Wasserstein metric. This space functions as the most general space of all distributions. We then construct

$$\mathcal{P}^\infty(\mathbb{R}^d) = \{\rho d^d x : \rho \in C^\infty(\mathbb{R}^d), \rho > 0, \int_{\mathbb{R}^d} \rho d^d x = 1\}, \quad (8)$$

the dense subspace of $\mathcal{P}_2(\mathbb{R}^d)$ of absolutely continuous measures with smooth positive density functions. This is the subspace we will principally work with and use as the ambient manifold from which we construct the submanifold of realizable distributions. While $\mathcal{P}_2(\mathbb{R}^d)$ is a formal ⁴ manifold, $\mathcal{P}^\infty(\mathbb{R}^d)$ is an actual smooth-infinite dimensional manifold with a smooth topology.

The tangent space at a sufficiently regular probability density q , which can be derived heuristically by manipulating the continuity equation and which we demonstrate more rigorously later, is identified with

$$T_{q d^d x} \mathcal{P}_2(\mathbb{R}^d) = {}^5 T_q \mathcal{P}_2(\mathbb{R}^d) = \left\{ \dot{q} = -\nabla \cdot (qv) : v \in L^2(q; \mathbb{R}^d) \right\}, \quad (9)$$

where v is a neural realizable velocity field⁶ or, equivalently, a vector field on the output space $\mathbb{R}^d$. For simplicity, we hereon refer to the functions given by this divergence operator, $\dot{q}$, as the **distributional deformations** induced by that velocity field, since the functions from the divergence indicate the instantaneous change in the distribution at each point and at each moment⁷. We further remark that all distributional deformations generated in this manner with a neural realizable velocity field must be mass-preserving; that is,

$$\int_{\mathbb{R}^d} \dot{q} d^d x = 0. \quad (10)$$

³For different neural architectures with different interpretations on their outputs, note that this can be generalized to any manifold. For example, for a feedforward neural network, setting $M = \mathbb{R}^d$, its output space, would be the reasonable choice. For a Transformer, as an example, it could be the probability simplex over the vocabulary, representing all distributions over next tokens.

⁴By formal, we mean that the geometric constructions are performed under the assumption that the Wasserstein space behaves as, but is not actually yet proven to be, a smooth infinite-dimensional manifold.

⁵For simplicity of notation, while technically each distribution, which is a point on $\mathcal{P}^\infty(\mathbb{R}^d)$, is followed by the volume form $d^d x$, or dvol_M more generally for any manifold M , in our paper we omit it such as here. In the future, we will generally omit writing $d^d x$ or dvol_M explicitly to avoid repetition, but the underlying object is still a probability measure.

⁶To briefly define, neural realizable simply means a velocity field that a given neural architecture can undergo via a finite parameter perturbation. Based upon our results in section 3, this notion can be made more rigorous by noting that it is simply the image of the neural tangent kernel of the neural function: $\mathcal{V} = \text{Im}(\Theta_\theta)$. Tentatively, we will first define it as $L^2(q; \mathbb{R}^d)$, the ambient Hilbert space of square-integrable velocity fields. In fact, $\mathcal{V} \subset L^2(q; \mathbb{R}^d)$. One may also observe that $\mathcal{V} \cong \mathbb{R}^p / \ker(J_\theta f_\theta)$ from the first isomorphism theorem.

⁷Lott’s construction asserts that the tangent space of points on $\mathcal{P}_2(\mathbb{R}^d)$ are given by scalar functions that differ by at most a constant. That is, $T_q \mathcal{P}_2(\mathbb{R}^d) \cong \mathbb{C}^\infty(\mathbb{R}^d)/\mathbb{R}$. This means that the velocity fields here can be given by the gradients of a family of functions that differ by at most a constant: $v = \nabla_x \phi = \nabla_x(\phi + c)$. See appendix section 9.2 for more discussion on this.

3 NTK and Neural Network Training via Drift Fields

Now, with a thorough introduction to all prior concepts, such as flow matching, drifting models, and with all the preliminary background established, let us now attempt to answer the question we posed in the introduction, formalized by equation 4: what is the relationship between the drift field and the effect of that training, quantified by the velocity field of the network?

Let us start again by expanding out the definition of the velocity field of a sample ϵ during neural network training:

$$u(f_\theta(\epsilon)) = \frac{d}{dt} f_{\theta(t)}(\epsilon) = J_\theta f_\theta \dot{\theta} \quad (11)$$

This basically means that the rate at which the output of the neural network changes can be calculated by the Jacobian of the time derivative of the parameters. We can think of the Jacobian as a pushforward from the space of instantaneous parameter changes to the output space.

Without loss of generality, let us suppose that this neural network is being trained by gradient descent under the gradient of some loss function. Gradient descent is essentially the Euler method applied to the continuous ODE involving the gradient of the loss function:

$$\dot{\theta}(t) = -\nabla_\theta L(\theta(t)) \quad (12)$$

Now, if the loss depends on sampled from a distribution ϵ' , we notice:

$$\nabla_\theta L = \mathbb{E}_{\epsilon'} [\nabla_\theta l(f_\theta(\epsilon'))] = \mathbb{E}_{\epsilon'} [J_\theta f_\theta(\epsilon')^\top \nabla_x l(f_\theta(\epsilon'))] \quad (13)$$

This is using the Jacobian of the neural network function as a pullback of the gradient of the loss in terms of the output to the parameter.

Then, substituting 13 into 12 to obtain ODE in terms of the gradient, and 12 into 11, we obtain an identity that relates the velocity field in terms of the gradient of the loss function with respect to the output space:

$$u_t(f_\theta(\epsilon)) = -J_\theta f_\theta(\epsilon) \mathbb{E}_{\epsilon'} [J_\theta f_\theta(\epsilon')^\top \nabla_x l(f_\theta(\epsilon'))] = -\mathbb{E}_{\epsilon'} [J_\theta f_\theta(\epsilon) J_\theta f_\theta(\epsilon')^\top \nabla_x l(f_\theta(\epsilon'))] \quad (14)$$

The crucial realization here is that the term $J_\theta f_\theta(\epsilon) J_\theta f_\theta(\epsilon')^\top$ is the definition of the **matrix-valued⁸ neural tangent kernel** (NTK) of a neural architecture, denoted by $\Theta_\theta : \mathbb{R}^m \times \mathbb{R}^m \rightarrow \mathbb{R}^{d \times d}$. To briefly introduce, the neural tangent kernel is a useful tool often employed to study the behavior of a neural network during training Jacot et al. [2018]. So, one can rewrite this into

$$u_t(f_\theta(\epsilon)) = -\mathbb{E}_{\epsilon'} [\Theta_\theta(\epsilon, \epsilon') \nabla_x l(f_\theta(\epsilon'))] \quad (15)$$

We now have an identity that relates the velocity field of the neural network undergoing training with the gradient of the loss function that is driving the training, in terms of the neural tangent kernel of that network. Conceptually, this can be interpreted as the neural tangent kernel "weighing" the relative effect of all other samples ϵ' on the gradient of the loss with respect to the output space in its effect on ϵ . This is already a coherent and general picture that reveals the general dynamic of how any neural network quantitatively changes under training by the gradient of a loss function.

Now, let us apply the specific type of loss function in this specific case to this identity we derived. When we use the drift field proposed by Deng et al. [2026], the loss function framed in terms of the drift field is given by

⁸There is a distinction between scalar-valued and matrix-valued neural tangent kernel. The former is more common in machine learning literature. The definitional difference is that the former is defined via the inner product between the two gradients, instead of the outer product.

$$\mathcal{L} = \mathbb{E}_{\epsilon}[\|f_{\theta}(\epsilon) - \text{stopgrad}(f_{\theta}(\epsilon) + \mathbf{V}_{p,q_{\theta}}(f_{\theta}(\epsilon)))\|^2] \quad (16)$$

Taking the gradient of this loss function with respect to the output x on the output space gives⁹

$$\nabla_x l(x) = -2\mathbf{V}_{p,q}(f_{\theta}(\epsilon)) \quad (17)$$

The multiplicative factor of 2 here can be entirely discarded by absorbing it under training step. So, let us substitute this gradient back to equation 15

$$u_t(f_{\theta}(\epsilon)) = \mathbb{E}_{\epsilon'}[\Theta_{\theta}(\epsilon, \epsilon')\mathbf{V}_{p,q}(f_{\theta}(\epsilon'))] \quad (18)$$

Expanding out the expectation into an integral,

$$u_t(f_{\theta}(\epsilon)) = \int_X \Theta_{\theta}(\epsilon, \epsilon')\mathbf{V}_{p,q_{\theta}}(f_{\theta}(\epsilon'))dp_{\text{prior}}(\epsilon') \quad (19)$$

We can then rewrite this more succinctly by treating the neural tangent kernel as an integral operator: $\Theta_{\theta} : L^2(p_{\text{prior}}; \mathbb{R}^d) \rightarrow L^2(p_{\text{prior}}; \mathbb{R}^d)$ ¹⁰:

Proposition 1 (NTK acts as a mediator between induced velocity field and drift field).

$$u_t(f_{\theta}(\epsilon)) = \Theta_{\theta}\mathbf{V}_{p,q_{\theta}}(f_{\theta}(\epsilon)) = \int_X \Theta_{\theta}(\epsilon, \epsilon')\mathbf{V}_{p,q_{\theta}}(f_{\theta}(\epsilon'))dp_{\text{prior}}(\epsilon')$$

Conceptually, this answers the previously unknown relation of Equation 4. This is the most important identity that enables our geometric construction. Let us unpack what this means: the actual velocity field that a sample undergoes in the output space is equal to the neural tangent kernel, in its integral operator form, associated with that neural network acting upon where the drift field intends to drift the network. One can imagine that the drift field is a raw force acting upon the network, pushing it towards desired, intended distribution. Yet this force is filtered by a kernel that is inherent and particular to that neural network. So, the geometry of the distribution evolution the neural network ultimately realizes is governed by the neural tangent kernel as a mediator. From this proposition it likewise naturally follows that the set of all neural realizable velocities are the image of the neural tangent kernel, the set of all vectors that it reaches: $\mathcal{V}_{\theta} = \text{Im}(\Theta_{\theta})$ ¹¹.

This paper takes this insight and seeks to give more rigorous theoretical grounding to it and discuss its implications. We intend to first show, using existing tools from relevant fields in mathematics, especially differential geometry, linear algebra, operator theory, functional analysis, that the distributions reachable by a neural architecture forms a Riemmanian manifold, and that the neural tangent kernel indeed gives rise to a geometric metric defined on that manifold. We then explore some implications of this geometric picture and discuss its applications. Under this framework, distribution evolution as a whole, a major topic in generative modeling, becomes naturally generalized into a curve on that manifold, where each point corresponds to one distribution. We then discuss an important application of this picture, which is that this picture enables comparison of the expressive power of neural architectures.

⁹The stopgrad operator here makes the term inside constant during differentiation. See the appendix of Deng et al. [2026], wherein they have a clearer derivation of this gradient.

¹⁰The neural tangent kernel can be viewed as a kernel function outputting one real vector when there are only two samples, a matrix when there are multiple samples, and, like this, an integral operator. Hereon in this paper the integral operator view takes precedence since it is the most general, but in most literature the single kernel view is the most common. In appendix 9.3, we also show that, from a functional analytical perspective, the drift field is a member of $L^2(p_{\text{prior}}; \mathbb{R}^d)$ under very mild assumptions.

¹¹Or, equivalently, since $\Theta_{\theta} = J_{\theta}f_{\theta}J_{\theta}f_{\theta}^T$, the image of the first Jacobian term: $\text{Im}(J_{\theta}f_{\theta})$. But the domain of this will be the parameter perturbation, not the drift field. But in both cases, the definition is equivalent: the set of all velocities a parameter perturbation or a drift field may induce.

4 NTK-Induced Riemannian Metric on the Parameter–Distribution Graph

4.1 Defining the Parameter-Distribution Graph from Distribution Locus

4.1.1 Defining Neural Architecture Induced Distributions via Probability Masses

Now that we understand the mediating role of the neural tangent kernel during neural network training, we now wish to expand this idea into the full. Before constructing the geometry of neural architecture realizable distributions, we must first rigorously establish that notion: *how does a neural architecture induce a distribution?* Let us revisit equation (1), the central finding of Deng et al. [2026]’s paper, that neural network induced functions operate as pushforward functional between two distributions: $q_t = [f_t]_{\#} p_{\text{prior}}, \forall t \in [T]$. What exactly does this mean? Let us make this notion more mathematically rigorous.

Suppose we take n samples from a prior distribution p_{prior} . We send these n samples through the same neural network parametrized by θ . This gives a set of points on the output space:

$$\{f_{\theta}(\epsilon_1), f_{\theta}(\epsilon_2), \dots, f_{\theta}(\epsilon_n)\} \subset \mathbb{R}^d, \text{ where } \epsilon_1, \dots, \epsilon_n \stackrel{i.i.d}{\sim} p_{\text{prior}} \quad (20)$$

An immediate interpretation that would emerge is that each of these outputs, $f_{\theta}(\epsilon_i)$, for all i , represents probability masses on the space $\mathbb{R}^d$. Together, they reveal the shape of the distribution that the neural network with this particular set of parameters, θ , would push samples to. The more samples there are, the more clearly it reveals the distribution on the output space. This is how a neural network acting as a pushforward functional between the prior distribution and current distribution given by θ operates.

Let us make this operational procedure mathematically rigorous. Since we have asserted that each one of these points on the output space function as probability masses, let us construct a probability distribution induced by these probability masses as follows. Let us define

$$\mu_n^{\theta}(p_{\text{prior}}) = \frac{1}{n} \sum_{i=1}^n \delta_{f_{\theta}(\epsilon_i)}, \epsilon_i \sim p_{\text{prior}} \quad (21)$$

where δ_x denotes the Dirac measure, which assigns the number 1 to the single point x . So, this is essentially a probability distribution based on an averaging of all n probability masses.

Then, immediately, the next move will be to take $n \rightarrow \infty$. That is, as the number of samples increases without bound, the probability mass will accumulate and eventually yield a probability density on the output space. This is precisely the notion of the pushforward as indicated in equation 1. We notice that, for any finite n , the distribution μ_n^{θ} is atomic¹², and thus certainly cannot be smooth. We observe and concede that there are also cases where even if the limit $n \rightarrow \infty$ is taken, the resulting distribution could fail to be smooth. But under mild assumptions on the input distribution, input dimension and the rank of the Jacobian of the neural architecture¹³, the resulting non-atomic distribution is strictly nonnegative and smooth, and thus is a member of $\mathcal{P}^{\infty}(\mathbb{R}^d)$. With this established, and further assuming that the prior distribution likewise is smooth and strictly positive, now we can say the pushforward can be viewed as an operator between smooth distributions on the input space to those on the output space:

$$(f_{\theta})_{\#} : \mathcal{P}^{\infty}(\mathbb{R}^m) \longrightarrow \mathcal{P}^{\infty}(\mathbb{R}^d) \quad (22)$$

being defined as

$$q_{\theta} = (f_{\theta})_{\#} = \lim_{n \rightarrow \infty} \mu_n^{\theta} \quad (23)$$

¹²An atomic probability distribution is a probability measure that assigns a positive, nonzero probability to at least one individual point, or an element in its event space. A smooth probability distribution is necessarily non-atomic, since it would have a probability density that each individual sample point can only have a value of 0.

¹³See appendix 9.2 for more discussion on this topic: the conditions under which neural architecture-generated distributions are smooth.

Now, the pushforward operator as introduced in the very beginning of this paper is made mathematically rigorous from a mere operational procedure of sampling and then individually sending forward.

With this established, let us now use this as a tool to induce the probability distribution space. Let us define the following: by having the parameter span over the entire parameter space, let us tentatively define the space of all reachable distributions by a particular neural architecture as

$$\mathcal{P}_f^{\mathcal{P}_{\text{prior}}} = \{(f_\theta)_\# p_{\text{prior}} \mid \theta \in \mathbb{R}^p\} \quad (24)$$

The most obvious observation here is that this is necessarily a subset of the space of all distributions, since certainly there are distributions that are not reachable. So right now we have this relation:

$$\mathbb{R}^p \xrightarrow{(f_\theta)_\#} \mathcal{P}_f^{\mathcal{P}_{\text{prior}}} \subset \mathcal{P}^\infty(\mathbb{R}^d) \quad (25)$$

It is important to note here that so far we have only shown that $\mathcal{P}_f^{\mathcal{P}_{\text{prior}}}$ is a space but have not imposed any additional geometric constraints.

Now, let us use some tools from differential geometry to additionally discuss what geometric properties such a space would have. We want to show that a subset of that space satisfies certain desirable geometric properties such that it qualifies locally as **immersion onto an ambient probability manifold**. To do this, we must first do some preliminary work on Lott's work as mentioned in section 2 of this paper, to show that the ambient space of smooth distributions is an infinite-dimensional manifold with an appropriate chart to a Hilbert space and Fréchet differentiability so as to employ the **Constant rank theorem** in an infinite-dimensional setting.

4.1.2 Regular Probability Space via Log Density Parametrization and Chart Construction to Hilbert Manifold

First, we must establish an ambient probability manifold from which immersions could be defined. Let us recall that Lott considers, for a smooth connected closed Riemannian manifold M , the space

$$\mathcal{P}^\infty(M) = \left\{ q \, d\text{vol}_M : q \in C^\infty(M), \, q > 0, \, \int_M q \, d\text{vol}_M = 1 \right\}, \quad (26)$$

equipped with the smooth topology. We have already established that this is a smooth manifold. The present setting differs in that the output space is $\mathbb{R}^d$, which is noncompact. In particular, a strictly positive probability density on $\mathbb{R}^d$ will generally satisfy

$$\inf_{x \in \mathbb{R}^d} q(x) = 0, \quad (27)$$

so positivity is not conveniently preserved by arbitrary small additive perturbations in the same manner as on a compact manifold. We therefore employ a relative log-density parametrization with a fixed reference probability density to establish charts on this ambient manifold, a common technique in information geometry and statistical manifolds that we borrow here.

Fix a smooth, strictly positive reference probability density

$$q_0 \in C^\infty(\mathbb{R}^d), \quad q_0(x) > 0, \quad \int_{\mathbb{R}^d} q_0(x) \, dx = 1. \quad (28)$$

For example, q_0 may be chosen to be the standard Gaussian distribution, since that is the simplest distribution easiest to sample from and with the most well known properties.

Let $s > d/2$ and consider the **Sobolev space** $H^s(\mathbb{R}^d)$ ¹⁴. By the Sobolev embedding theorem,

$$H^s(\mathbb{R}^d) \hookrightarrow L^\infty(\mathbb{R}^d). \quad (29)$$

¹⁴A Sobolev space $W^{k,p}$ over an open set $\Omega \subseteq \mathbb{R}^p$ is defined as the subset of functions in $L^p(\Omega)$ such that f and its weak derivatives up to order k have a finite L^p norm. When $p = 2$, the Sobolev space is a Hilbert space equipped with an inner product and is denoted as H^s .

Define the closed codimension-one Hilbert subspace. This space will serve as tangent directions of deformations that serve as distribution coordinates.

$$H_{0,q_0}^s(\mathbb{R}^d) := \left\{ h \in H^s(\mathbb{R}^d) : \int_{\mathbb{R}^d} h(x) q_0(x) dx = 0 \right\}. \quad (30)$$

For $h \in H_{0,q_0}^s(\mathbb{R}^d)$, define

$$Z(h) := \int_{\mathbb{R}^d} e^{h(x)} q_0(x) dx \quad (31)$$

and

$$\Psi_{q_0}(h)(x) := \frac{e^{h(x)} q_0(x)}{Z(h)}. \quad (32)$$

Since $s > d/2$, h is bounded. Hence

$$0 < e^{-\|h\|_\infty} \leq e^{h(x)} \leq e^{\|h\|_\infty}, \quad (33)$$

and consequently

$$0 < Z(h) < \infty. \quad (34)$$

It follows immediately from Eq. (32) that

$$\Psi_{q_0}(h)(x) > 0 \quad (35)$$

for all $x \in \mathbb{R}^d$, and

$$\int_{\mathbb{R}^d} \Psi_{q_0}(h)(x) dx = 1. \quad (36)$$

Thus positivity and normalization are built directly into the parametrization.

The normalization in Eq. (32) introduces an additive redundancy, since for every constant $C \in \mathbb{R}$,

$$\frac{e^{h+C} q_0}{\int e^{h+C} q_0 dx} = \frac{e^h q_0}{\int e^h q_0 dx}. \quad (37)$$

Condition (30) fixes this redundancy by selecting a unique representative.

We therefore define the regular density class, again fixing a reference q_0 :

$$\mathcal{P}_{q_0}^s(\mathbb{R}^d) := \left\{ q : q > 0, \int_{\mathbb{R}^d} q dx = 1, \log \left(\frac{q}{q_0} \right) \in H^s(\mathbb{R}^d) + \mathbb{R} \right\}. \quad (38)$$

For $q \in \mathcal{P}_{q_0}^s(\mathbb{R}^d)$, define

$$\chi_{q_0}(q) := \log \left(\frac{q}{q_0} \right) - \int_{\mathbb{R}^d} \log \left(\frac{q(x)}{q_0(x)} \right) q_0(x) dx. \quad (39)$$

By construction,

$$\chi_{q_0}(q) \in H_{0,q_0}^s(\mathbb{R}^d). \quad (40)$$

Moreover,

$$\chi_{q_0} \circ \Psi_{q_0} = \text{Id}_{H_{0,q_0}^s}, \quad (41)$$

and

$$\Psi_{q_0} \circ \chi_{q_0} = \text{Id}_{\mathcal{P}_{q_0}^s}. \quad (42)$$

Consequently,

$$\chi_{q_0} : \mathcal{P}_{q_0}^s(\mathbb{R}^d) \longrightarrow H_{0,q_0}^s(\mathbb{R}^d) \quad (43)$$

provides a Hilbert-space coordinate representation of the regular density class reflecting the tangent directions acting the chosen prior distribution, with inverse Ψ_{q_0} , to a Hilbert manifold. We now have a usable ambient probability space equipped with a smooth structure.

4.1.3 Derivation of the Candidate Differential of Pushforward

Before we use the Constant Rank Theorem, we must first discuss the differential of the chart we derived above. The differential of the inverse chart is also explicit. Let $\xi \in H_{0,q_0}^s(\mathbb{R}^d)$. Differentiating Eq. (32) gives

$$D\Psi_{q_0}(h)[\xi] = q_h \left(\xi - \int_{\mathbb{R}^d} \xi(x) q_h(x) dx \right), \quad (44)$$

where

$$q_h := \Psi_{q_0}(h). \quad (45)$$

In particular, at $h = 0$, where $q_h = q_0$, the zero-mean condition implies

$$D\Psi_{q_0}(0)[\xi] = q_0 \xi. \quad (46)$$

Furthermore,

$$\int_{\mathbb{R}^d} D\Psi_{q_0}(h)[\xi] dx = 0, \quad (47)$$

so tangent vectors obtained through the chart are precisely infinitesimal mass-preserving density deformations. This matches our earlier observation that the integral over the entire space of distributional deformations must be 0. This has the interpretation of preserving the mass the probability density at 1 when undergoing that deformation.

We now apply this chart to the architecture-generated distributions. Let

$$\Phi : \mathbb{R}_{\text{reg}}^p \longrightarrow \mathcal{P}_{q_0}^s(\mathbb{R}^d), \quad \Phi(\theta) = q_\theta = (f_\theta)_\# p_{\text{prior}}, \quad (48)$$

where the regularity assumptions on f_θ and p_{prior} ensure that q_θ belongs to the density class in Eq. (38). In the chart χ_{q_0} , the architecture map becomes

$$\widehat{\Phi} := \chi_{q_0} \circ \Phi : \mathbb{R}_{\text{reg}}^p \longrightarrow H_{0,q_0}^s(\mathbb{R}^d), \quad (49)$$

with

$$\widehat{\Phi}(\theta) = \log \left(\frac{q_\theta}{q_0} \right) - \int_{\mathbb{R}^d} \log \left(\frac{q_\theta(x)}{q_0(x)} \right) q_0(x) dx. \quad (50)$$

The differentiability question for Φ may therefore be formulated as an ordinary Fréchet differentiability problem between the finite-dimensional regular parameter space and the Hilbert model space. Explicitly, $\widehat{\Phi}$ is Fréchet differentiable at θ if there exists a bounded linear map

$$D\widehat{\Phi}_\theta : \mathbb{R}_{\text{reg}}^p \longrightarrow H_{0,q_0}^s(\mathbb{R}^d) \quad (51)$$

such that

$$\lim_{\|h\| \rightarrow 0} \frac{\left\| \widehat{\Phi}(\theta + h) - \widehat{\Phi}(\theta) - D\widehat{\Phi}_\theta[h] \right\|_{H^s}}{\|h\|_{\mathbb{R}^p}} = 0. \quad (52)$$

To prove Fréchet differentiability of $\widehat{\Phi}$, we need to first discuss differentiability of Φ . Independently of the choice of chart, differentiation of the pushforward identity gives the intrinsic distributional differential. For any smooth test function φ ,

$$\int_{\mathbb{R}^d} \varphi(x) q_\theta(x) dx = \int \varphi(f_\theta(\epsilon)) dp_{\text{prior}}(\epsilon). \quad (53)$$

Differentiating in the direction $\dot{\theta} \in \mathbb{R}^p$ gives

$$\frac{d}{dt} \bigg|_{t=0} \int_{\mathbb{R}^d} \varphi(x) q_{\theta+t\dot{\theta}}(x) dx = \int \nabla \varphi(f_\theta(\epsilon))^\top J_\theta f_\theta(\epsilon) \dot{\theta} dp_{\text{prior}}(\epsilon). \quad (54)$$

Writing the architecture-realizable velocity field as

$$u_{\dot{\theta}}(f_\theta(\epsilon)) = J_\theta f_\theta(\epsilon) \dot{\theta}, \quad (55)$$

and pushing the integral forward to output space yields

$$\int_{\mathbb{R}^d} \nabla \varphi(x)^\top u_{\dot{\theta}}(x) q_{\theta}(x) dx. \quad (56)$$

Integration by parts therefore gives

$$\int_{\mathbb{R}^d} \varphi(x) \cdot [-\nabla \cdot (q_{\theta} u_{\dot{\theta}})] dx \quad (57)$$

Equating this with the left hand side of 54 again:

$$\left. \frac{d}{dt} \right|_{t=0} \int_{\mathbb{R}^d} \varphi(x) q_{\theta+t\dot{\theta}}(x) dx = \int_{\mathbb{R}^d} \varphi(x) \cdot [-\nabla \cdot (q_{\theta} u_{\dot{\theta}})] dx \quad (58)$$

And realize the the time derivative precisely gives the distributional deformation, $\frac{d}{dt} q_{\theta+t\dot{\theta}} = \dot{q}$, so

$$\int_{\mathbb{R}^d} \varphi(x) \dot{q} dx = \int_{\mathbb{R}^d} \varphi(x) \cdot [-\nabla \cdot (q_{\theta} u_{\dot{\theta}})] dx \quad (59)$$

Since φ is an arbitrary test function, the above equation of both integrals would yield

$$\boxed{\dot{q} = d\Phi_{\theta}(\dot{\theta}) = -\nabla \cdot (q_{\theta} u_{\dot{\theta}}) = -\nabla \cdot (q_{\theta} J_{\theta} f_{\theta} \dot{\theta})}. \quad (60)$$

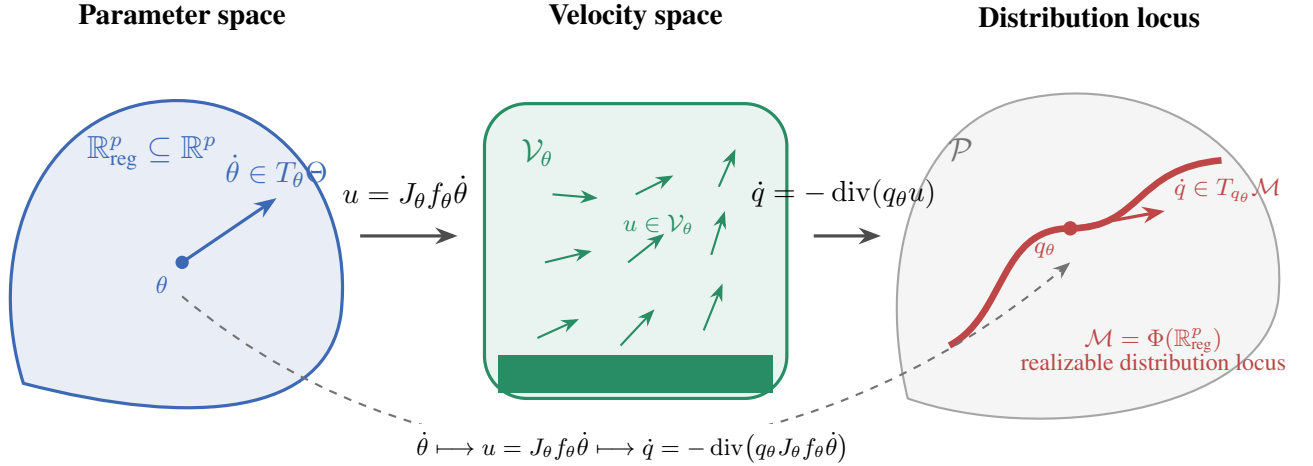


Figure 2: A parameter-space perturbation $\dot{\theta}$ induces the velocity field $u = J_{\theta} f_{\theta} \dot{\theta}$, which in turn induces the distributional tangent direction $\dot{q} = -\text{div}(q_{\theta} u)$ along the realizable distribution locus $\mathcal{M} = \Phi(\Theta)$. This captures the three-step process that undergirds architecture generated distributional change.

Note that this is exactly what we have asserted in section (2), that the tangent space of the ambient probability manifold is the set of all neural realizable distributional deformations given by the negative divergence of the current distribution weighed by the velocity field acting on it. Also note that this identity can also be obtained heuristically simply via the continuity equation by simply moving the divergence term to the other side:

$$\frac{d}{dt} q_{\theta(t)} + \nabla \cdot (q_{\theta(t)} u) = 0 \quad (61)$$

Thus,

$$\frac{d}{dt}q_{\theta(t)} = -\nabla \cdot (q_{\theta(t)}u) = -\nabla \cdot (q_{\theta(t)}J_{\theta}f_{\theta}\dot{\theta}) \quad (62)$$

Let us also remark that this differential has a very clear conceptual interpretation, aside from its mathematical significance: it is a pushforward from a parameter perturbation $\dot{\theta}$, which belongs to $T\mathbb{R}^p \cong \mathbb{R}^p$, to the distributional deformation, $\dot{q} = -\nabla \cdot (q_{\theta}J_{\theta}f_{\theta}\dot{\theta})$ that that perturbation in the parameter of the underlying neural architecture induces via the induced velocity as an intermediary:

$$\dot{\theta} \longrightarrow u = J_{\theta}f_{\theta}\dot{\theta} \longrightarrow \dot{q} = -\nabla \cdot (q_{\theta}u) \quad (63)$$

4.1.4 Using Constant Rank Theorem to Induce the Local Immersed Branches

With all previous work establishing charts to the Hilbert manifold and a discussion of the candidate differential, and with Fréchet differentiability of the mapping $\widehat{\Phi}$ proven¹⁵, the infinite-dimensional Constant Rank Theorem may finally be applied in the Hilbert-manifold setting. This is the crucial conceptual step that bridges the architecture realizable distributions with the probability space we constructed. We first define the **regular parameter space**, $\mathbb{R}_{\text{reg}}^p \subset \mathbb{R}^p$, where the following differential is injective, since that is an assumption that the Constant Rank Theorem requires:

$$d\Phi_{\theta} : T_{\theta}\mathbb{R}_{\text{reg}}^p \cong \mathbb{R}^p \longrightarrow T_{q_{\theta}}\mathcal{P}_{q_0}^s(\mathbb{R}^d) \quad (64)$$

Then

$$\ker(d\Phi_{\theta}) = \{0\}, \quad (65)$$

which is trivially complemented. Moreover,

$$\dim \text{Range}(d\Phi_{\theta}) = p. \quad (66)$$

Since every finite-dimensional subspace of a Hilbert space is closed and complemented,

$$\text{Range}(d\Phi_{\theta}) \quad (67)$$

is automatically a closed complemented subspace of the ambient tangent space. Thus, provided the split-rank condition persists locally, and that the regular parameter space, as a subset of the total Euclidean parameter space $\mathbb{R}^p$, is open¹⁶, the Hilbert-manifold Constant Rank Theorem gives local coordinates in which Φ takes the normal form

$$\theta \longmapsto (\theta, 0). \quad (68)$$

For every $\theta \in \mathbb{R}_{\text{reg}}^p$, the Constant Rank Theorem provides an open neighborhood $U_{\theta} \subset \mathbb{R}_{\text{reg}}^p$ such that $\Phi_{U_{\theta}}$ is a smooth embedding. Hence, $\Phi(U_{\theta})$ is a p -dimensional embedded submanifold of $\mathcal{P}_{q_0}^s(\mathbb{R}^d)$. The global image of $\Phi(\mathbb{R}_{\text{reg}}^p)$, however, need not be an embedded submanifold because images arising from distinct parameter neighborhoods may intersect. To illustrate this principle, consider the classic example of figure eight in the $\mathbb{R}^2$ space: it is an immersion into $\mathbb{R}^2$, but it cannot be considered a submanifold of $\mathbb{R}^2$ since the point where it intersects would not be locally Euclidean. But the geometric picture that neural architecture induced distributions is clearly emerging. We need one more workaround to establish a global manifold that can encompass all regular and reachable distributions.

Let us tentatively define the **architecture-realizable distribution locus** as the union of all such embedded submanifolds, as follows:

¹⁵See appendix 9.5 for a full proof. We originally placed this as a subsection before this one but relegated it to the appendix due to its great length.

¹⁶See appendix 9.6 for proof and discussion of this. Openness of the subset is a requirement for the preimage in the Constant Rank Theorem, since it guarantees that every point has a neighborhood such that the differential could exist. The proof for both of these is in fact conceptually related.

Definition 2. (*Architecture-realizable distribution locus*) Let $\Phi : \mathbb{R}_{reg}^p \longrightarrow \mathcal{P}_{q_0}^s(\mathbb{R}^d)$ be the pushforward operator associated with the regular parameter space of a neural architecture defined as $\Phi(\theta) := (f_\theta)_{\#} p_{prior} = \lim_{n \rightarrow \infty} \mu_n^\theta$. We define

$$\mathfrak{M}_f^{p_{prior}} = \bigcup_{\theta \in \mathbb{R}_{reg}^p} \Phi(U_\theta),$$

which is equivalent to $\Phi(\mathbb{R}_{reg}^p)$. It is regarded as a subset of the ambient probability space and need not be a manifold.

It is easy to demonstrate that this locus inherits subspace topology of the regular density class $\mathcal{P}_{q_0}^s(\mathbb{R}^d)$. Although it need not be a manifold that is locally Euclidean over all points, it is a valid topological space.

Proposition 3. (*Topology of the architecture-realizable distribution locus*). $\mathfrak{M}_f^{p_{prior}}$ as defined above is a topological space, equipped with the subspace topology inherited from the ambient probability manifold $\mathcal{P}_{q_0}^s(\mathbb{R}^d)$.

Proof. Let $\tau_{\mathcal{P}}$ denote the topology of the ambient Hilbert manifold $\mathcal{P}_{q_0}^s(\mathbb{R}^d)$. Since

$$\mathfrak{M}_f^{p_{prior}} \subseteq \mathcal{P}_{q_0}^s(\mathbb{R}^d),$$

define

$$\tau_{\mathfrak{M}} := \left\{ O \cap \mathfrak{M}_f^{p_{prior}} \mid O \in \tau_{\mathcal{P}} \right\}.$$

We verify that $\tau_{\mathfrak{M}}$ is a topology. First,

$$\emptyset = \emptyset \cap \mathfrak{M}_f^{p_{prior}} \in \tau_{\mathfrak{M}}, \quad \mathfrak{M}_f^{p_{prior}} = \mathcal{P}_{q_0}^s(\mathbb{R}^d) \cap \mathfrak{M}_f^{p_{prior}} \in \tau_{\mathfrak{M}}.$$

For any family $\{O_\alpha\}_{\alpha \in A} \subseteq \tau_{\mathcal{P}}$,

$$\bigcup_{\alpha \in A} (O_\alpha \cap \mathfrak{M}_f^{p_{prior}}) = \left(\bigcup_{\alpha \in A} O_\alpha \right) \cap \mathfrak{M}_f^{p_{prior}} \in \tau_{\mathfrak{M}}.$$

Likewise, for $O_1, O_2 \in \tau_{\mathcal{P}}$,

$$(O_1 \cap \mathfrak{M}_f^{p_{prior}}) \cap (O_2 \cap \mathfrak{M}_f^{p_{prior}}) = (O_1 \cap O_2) \cap \mathfrak{M}_f^{p_{prior}} \in \tau_{\mathfrak{M}}.$$

Thus $\tau_{\mathfrak{M}}$ satisfies the axioms of a topology. Therefore,

$$\left(\mathfrak{M}_f^{p_{prior}}, \tau_{\mathfrak{M}} \right)$$

is a topological space. Moreover, the inclusion

$$j : \mathfrak{M}_f^{p_{prior}} \hookrightarrow \mathcal{P}_{q_0}^s(\mathbb{R}^d)$$

is continuous by construction. Since the ambient Hilbert manifold is Hausdorff, its subspace $\mathfrak{M}_f^{p_{prior}}$ is also Hausdorff. If the ambient manifold is assumed to be second-countable, the locus inherits that property as well.

□

Remark (Topological space versus topological manifold). This proposition does not assert that $\mathcal{M}_f^{p_{\text{prior}}}$ is a topological manifold. Every subset of a topological space inherits a subspace topology, but a topological manifold must additionally be locally Euclidean. When distinct local branches $\Phi(U_{\theta_1})$ and $\Phi(U_{\theta_2})$ intersect, a neighborhood of their intersection may contain several incompatible branches and therefore need not be homeomorphic to an open subset of $\mathbb{R}^p$. The locus consequently remains a well-defined topological space even where its manifold structure fails.

Another remark that is due is related to the earlier space we induced, $\mathcal{P}_f^{p_{\text{prior}}}$. This is the set of *all* distributions a neural architecture can reach when we no longer restrict ourselves to the regular parameter space but instead have it vary across the entire Euclidean parameter space. We likewise remark here that, using the same logic as above, $\mathcal{P}_f^{p_{\text{prior}}}$ likewise is a topological space that inherits a subset topology, but this time from $\mathcal{P}_2$ equipped with the Wasserstein topology, since we cannot guarantee that all distributions reached are admissible by the regular probability space. Its geometric properties will necessarily be much less guaranteed than the distribution locus $\mathcal{M}_f^{p_{\text{prior}}}$. Since we no longer are confined to the regular parameter space, this will be a topological space with sudden rank changes, dimension changes from neighborhood to neighborhood, and other degenerate and less controlled behavior.

4.1.5 Constructing the Graph Submanifold of the Product Manifold

Now, we have tentatively induced the sets of local manifolds of realizable distributions $\mathcal{M}_f^{p_{\text{prior}}}$ using the regular parameter space, a subset of the parameter space of the neural architecture f . We likewise have shown some preliminary results on its geometric property, that it is a topological space inheriting a subspace topology. But, as we have established, this is not yet a global manifold immersed into the probability space. Furthermore, this construction alone is not sufficient for a Riemannian metric to be defined on it. In essence, the reason why $\mathcal{M}$ can neither be guaranteed to be manifold nor equipped with a metric is essentially the same: intersection points when multiple parameters map to the same distribution. Let us elaborate on the nature of this problem and discuss a crucial workaround that proves to be the conceptual center of this paper.

We have already established that the space $\mathcal{M}$ is constructed via a pushforward operator, as we have already mentioned, $\Phi : \mathbb{R}_{\text{reg}}^p \subset \mathbb{R}^p \longrightarrow \mathcal{M}_f^{p_{\text{prior}}}$. Note that, once we are pushed forward to the distributional manifold, we lose track of which parameter we came from that gave rise to a distribution. In other words, when we have $q_\theta \in \mathcal{M}_f^{p_{\text{prior}}}$, we cannot easily recover $\theta \in \mathbb{R}_{\text{reg}}^p$ for the simple reason that multiple parameters can give rise to the same distribution via Φ . Although we have stipulated that the *differential* of Φ is injective on the regular parameter space, it does not follow that the mapping itself is injective. So, this means it is possible that for $\theta_1 \neq \theta_2$, $\Phi(\theta_1) = \Phi(\theta_2)$. Looking at this from the perspective of the configuration of a neural network, this is likewise easy to see: especially for an overparametrized neural network, where $p \gg d$, a neural network can easily have parameter redundancies and symmetries such that different parameters give rise to the same function, and thus the same distribution. Or, even if the function is not the same pointwise, the distribution in the end is the same.

Why is this a problem? This foreshadows our construction of the Riemannian metric later. Since we see the manifold to be defined from a finite parameter space to the distribution space, when we pullback a metric from the distribution space, we must ensure the *identifiability* of the parameter it came from. Suppose if two different parameters give rise to the same distribution, we must then ensure that

$$g_{q_{\theta_1}}(\dot{q}_1, \dot{q}_2) = g_{q_{\theta_2}}(\dot{q}_1, \dot{q}_2)$$

But, this cannot be guaranteed. If it is possible that two different parameters give rise to the same distribution, then the local geometry and curvature nearby is almost certainly different. In that case, the metric defined onto the two distributions will not agree. But since we do not have parameter identifiability, once we push forward to the distribution space, we will lose track of which parameter we came from.

There are several solutions to this problem ¹⁷, but this paper adopts the following: using the graph induced subset of a **product manifold** between the distribution manifold and the regular parameter space. The concept of a product manifold is not difficult: it is simply the Cartesian product between two manifolds. In differential geometry, it is long established that the product manifold between two manifolds is also a manifold:

$$M \times N = \{(p_1, p_2) \mid p_1 \in M \text{ and } p_2 \in N\}, \quad (69)$$

where $\dim(M \times N) = \dim(M) + \dim(N)$. The benefit of using a product manifold in this situation is that each point can now encode both the distribution q_θ and the parameter θ that gives rise to it, which we can now see as a tuple. We hereon refer to such a tuple as a **parameter-distribution pair**.

But, we do not blindly use the entire product manifold, as there would be too many distribution-parameter pairs that are unrelated. So, here we adopt a specific construction called **graphs** of mapping. The graph of a mapping between two sets is a subset of their Cartesian product defined as follows: let $F : X \rightarrow Y$ be any mapping. The graph of F is the subset of the Cartesian product $X \times Y$ defined by

$$\text{Graph}(F) := \{(x, F(x)) \mid x \in X\} \subseteq X \times Y \quad (70)$$

Thus, using the graph of a mapping can extract all pairs that are related, obtaining the subset of the Cartesian product that is only of interest. Another remark is that, if X is a space endowed with dimensions, $\dim(\text{Graph}(F)) = \dim(X)$, and is not the sum of the dimensions of both spaces. This property also proves to be very beneficial since it preserves the dimensionality of the manifold we constructed.

Now, with this mathematical tool introduced, let us make the following definition.

Definition 4 (Graph-induced Submanifold of Parameter-Distribution Product Manifold). *Let $\Phi : \mathbb{R}_{\text{reg}}^p \rightarrow \mathcal{P}_{q_0}^s(\mathbb{R}^d)$ via $\Phi(\theta) = (f_\theta)_{\#} p_{\text{prior}}$. We define the **parameter-distribution graph manifold** of the realizable distributions by neural architecture-associated pushforward Φ as*

$$\Gamma_\Phi := \text{Graph}(\Phi) = \{(\theta, q_\theta) \mid \theta \in \mathbb{R}_{\text{reg}}^p\} \subset \mathbb{R}_{\text{reg}}^p \times \mathcal{P}_{q_0}^s(\mathbb{R}^d)$$

equipped with the graph map

$$\iota_\Phi : \mathbb{R}_{\text{reg}}^p \longrightarrow \mathbb{R}_{\text{reg}}^p \times \mathcal{P}_{q_0}^s(\mathbb{R}^d), \quad \iota(\theta) = (\theta, q_\theta)$$

Remark: The differential of the graph embedding will carry over easily as $d(\iota_\Phi)_\theta(\dot{\theta}) = (\dot{\theta}, d\Phi_\theta(\dot{\theta}))$. Thus, the tangent space of a point $(\theta, q_\theta) \in \Gamma_\Phi$ will be defined by $T_{(\theta, q_\theta)}\Gamma_\Phi = \{(\dot{\theta}, -\nabla \cdot (q_\theta J_\theta f_\theta \dot{\theta})) \mid \dot{\theta} \in T_\theta \mathbb{R}_{\text{reg}}^p\}$.

For simplicity, we hereon denote the tuple as bolded $\mathbf{q}_\theta := (\theta, q_\theta) \in \Gamma_\Phi$ and similarly $\dot{\mathbf{q}}_\theta = (\dot{\theta}, \dot{q}_\theta) \in T_{\mathbf{q}}\Gamma_\Phi$.

Proposition 5 (The graph map ι is an embedding).

Proof. Define the graph map

$$\iota_\Phi : \mathbb{R}_{\text{reg}}^p \longrightarrow \mathbb{R}_{\text{reg}}^p \times \mathcal{P}_{q_0}^s(\mathbb{R}^d), \quad \iota_\Phi(\theta) = (\theta, \Phi(\theta)).$$

The map ι_Φ is smooth because both the identity map on $\mathbb{R}_{\text{reg}}^p$ and Φ are smooth. It is also injective, since

$$\iota_\Phi(\theta_1) = \iota_\Phi(\theta_2)$$

implies equality of the first coordinates and hence

$$\theta_1 = \theta_2.$$

For any $\dot{\theta} \in T_\theta \mathbb{R}_{\text{reg}}^p$, the differential of ι_Φ is

$$d(\iota_\Phi)_\theta[\dot{\theta}] = \left(\dot{\theta}, d\Phi_\theta[\dot{\theta}] \right).$$

¹⁷See future work for more discussion on this.

Therefore,

$$d(\iota_\Phi)_\theta[\dot{\theta}] = (0, 0) \implies \dot{\theta} = 0,$$

because the first component of $d(\iota_\Phi)_\theta[\dot{\theta}]$ is $\dot{\theta}$. Thus $d(\iota_\Phi)_\theta$ is injective at every $\theta \in \mathbb{R}_{\text{reg}}^p$, so ι_Φ is an immersion.

The inverse of ι_Φ on its image Γ_Φ is the restriction of the first-coordinate projection:

$$\iota_\Phi^{-1} = \text{pr}_1|_{\Gamma_\Phi} : \Gamma_\Phi \longrightarrow \mathbb{R}_{\text{reg}}^p, \quad \text{pr}_1|_{\Gamma_\Phi}(\theta, \Phi(\theta)) = \theta.$$

Since coordinate projections on product manifolds are smooth, $\text{pr}_1|_{\Gamma_\Phi}$ is smooth. Consequently, ι_Φ is a smooth embedding, and its image Γ_Φ is an embedded submanifold of

$$\mathbb{R}_{\text{reg}}^p \times \mathcal{P}_{q_0}^s(\mathbb{R}^d).$$

Moreover, ι_Φ is a diffeomorphism from $\mathbb{R}_{\text{reg}}^p$ onto Γ_Φ , so

$$\Gamma_\Phi \cong \mathbb{R}_{\text{reg}}^p$$

and thus

$$\dim \Gamma_\Phi = \dim \mathbb{R}_{\text{reg}}^p = p$$

□

This proves that the graph is an embedded manifold in the product of the regular parameter space and the ambient distribution manifold of finite dimension p . This crucial mathematical workaround circumvents both the problem of the locus not being globally a manifold and a metric impossible to be defined thereon. We thus now have a valid global manifold that characterizes the regular distributions a neural architecture may realize in a subset of its parameter space. We consider this graph to be the smooth manifold of neural architecture realizable distribution.

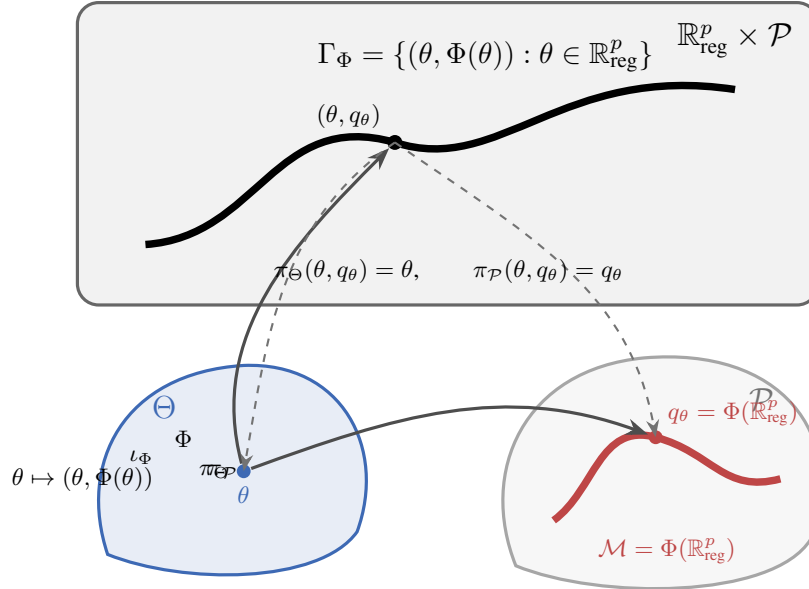


Figure 3: The map $\Phi : \mathbb{R}_{\text{reg}}^p \rightarrow \mathcal{P}$ determines both the realizable distribution locus $\mathcal{M} = \Phi(\mathbb{R}_{\text{reg}}^p)$ and its graph $\Gamma_\Phi \subseteq \mathbb{R}_{\text{reg}}^p \times \mathcal{P}$. The graph embedding $\iota_\Phi(\theta) = (\theta, \Phi(\theta))$ retains the parameter associated with each distribution, whereas projection onto $\mathcal{P}$ recovers the locus: $\pi_\mathcal{P}(\Gamma_\Phi) = \mathcal{M}$.

4.2 Using NTK to Induce Riemannian Metric on the Parameter-Distribution Graph

4.2.1 Onsager Interpretation of NTK as Mobility Tensor

We have so far shown that there does exist a family of local spaces of realizable distributions. We also have constructed a graph manifold to pair the distribution locus with the parameter space giving rise to it. But the ultimate question we wish to answer still is not resolved: what would be its metric? What is the relation between its geometry and the neural architecture that gave rise to it? To appropriately answer this question, we must introduce certain tools from other conceptual areas, especially the connection between Onsager geometry and Riemannian geometry. This is where the identity we derived long back in section 3, that the NTK mediates an external drift field with actual realized velocity becomes relevant again.

Let us state the continuity equation again, this time defined on the parameter-distribution graph:

$$\frac{\partial}{\partial t} \mathbf{q}_t + \nabla \cdot (\mathbf{q}_t u_t) = 0 \quad (71)$$

Substituting from equation (10), the aforementioned identity $u = \Theta V$, into the velocity field, we can rewrite this as

$$\frac{\partial}{\partial t} \mathbf{q}_t + \nabla \cdot (\mathbf{q}_t \Theta_\theta \mathbf{V}_{p,q_t}) = 0 \quad (72)$$

Let us make an observation on this: recall that in our discussion on Wasserstein gradient flow, the relation we had is that u_t , the velocity field, is directly given by the negative gradient of the variational derivative of the discrepancy functional. Recall:

$$u_t = -\nabla \frac{\delta D}{\delta \mathbf{q}_t} \quad (73)$$

In other words, in the standard Wasserstein formulation, the spatial velocity is the negative spatial gradient of the variational derivative. The conversion from this velocity to a density deformation is encoded by the continuity equation.

$$\frac{\partial}{\partial t} \mathbf{q}_t = \nabla \cdot (\mathbf{q}_t I \nabla \frac{\delta D}{\delta \mathbf{q}_t}) \quad (74)$$

One can think of this analogously using physics: the negative gradient of the discrepancy is the "force" that is driving the velocity field. In the case of Wasserstein gradient flow, the force is able to directly map onto the velocity. But that is not the case in all physical or dynamical systems.

This is when Onsager geometry provides the theoretical foundation for what we are discussing. Onsager geometry posits that the more general relation is given by

$$u = -M(\mathbf{q}) \nabla \frac{\delta D}{\delta \mathbf{q}} \quad (75)$$

where $M(\mathbf{q})$, the general quantity that mediates the induced velocity from the force, is called the **mobility tensor**. This framework thus generalizes our earlier discussion on Wasserstein gradient flows, since that would be the special case where $M(\mathbf{q}) = I$.

Now that we have this general relation, let us go back to what we already derived: from the continuity equation given in equation (25), we notice that $M(\mathbf{q})$, the mobility tensor is precisely a role taken by the neural tangent kernel, since the neural tangent kernel is the term that acts upon the drift field, the force:

$$u_t = \Theta_\theta \mathbf{V}_{p,q_t} \implies M_\theta = \Theta_\theta \quad (76)$$

The NTK therefore plays the role of a mobility operator on latent output space, analogous to the force-to-velocity operator in an overdamped system with linear resistance. We have noticed this heuristically from the observation that it matches the form $v = MF$. But let us prove it more rigorously here.

Theorem 6 (The NTK induces a mobility on latent output space). *The neural tangent kernel integral operator*

$$\Theta_\theta = A_\theta A_\theta^* : \mathcal{H} \longrightarrow \mathcal{H}$$

defines a positive-semidefinite mobility

$$M_\theta := \Theta_\theta \mathcal{R}_\mathcal{H}^{-1} : \mathcal{H}^* \longrightarrow \mathcal{V}_\theta.$$

Moreover, its restriction induces a positive-definite mobility

$$M_\theta^\mathcal{V} := (\Theta_\theta|_{\mathcal{V}_\theta}) \mathcal{R}_{\mathcal{V}_\theta}^{-1} : \mathcal{V}_\theta^* \longrightarrow \mathcal{V}_\theta,$$

where $\mathcal{R}_{\mathcal{V}_\theta}$ is the Riesz isomorphism for the inherited Hilbert inner product.

Thus the NTK converts force covectors, represented by elements of $\mathcal{H}$, into realizable latent velocity fields.

To prove this observation, we must first specify the domain and codomain of the integral operator. As discussed in section 3, we have already established that drift fields and velocity fields are both members of the L^p space in functional analysis $L^2(p_{\text{prior}}; \mathbb{R}^d) = \{f : \mathbb{R}^d \longrightarrow \mathbb{R}^d \mid \int_{\mathbb{R}^d} |f(x)|^2 p_{\text{prior}}(x) < \infty\}$. More specifically, we have discussed how neural realizable velocity fields, $\mathcal{V}_\theta$, must be a subset of this space. Here, we take advantage of an important result from functional analysis, that L^p spaces are **Hilbert spaces** when $p = 2$. By the Riesz representation theorem, there exists an isometric isomorphism $\mathcal{R}_H : H \longrightarrow H^*$.

The relevance of this result is that forces are conceptually identified as **covectors** and velocities are identified as tangent vectors. So while the domain and codomain of Θ_θ appears to be the same, we must conceptually recognize they are different and interpret the domain instead as the dual space of the L^2 space¹⁸. So, we first posit and identify that

$$\Theta_\theta : \mathcal{H}^* \cong \mathcal{H} \longrightarrow \mathcal{H} \tag{77}$$

With this crucial identity from functional analysis discussed, let

$$\mathcal{H} := L^2(p_{\text{prior}}; \mathbb{R}^d), \quad \langle v, w \rangle_{\mathcal{H}} := \int_X v(\epsilon)^\top w(\epsilon) p_{\text{prior}}(\epsilon).$$

Let us further revisit the definition of the matrix-valued neural tangent kernel:

$$J_\theta(\epsilon) := D_\theta f_\theta(\epsilon) \in \mathbb{R}^{d \times p},$$

and assume that

$$\int_X \|J_\theta(\epsilon)\|_F^2 p_{\text{prior}}(\epsilon) < \infty.$$

Define the matrix-valued neural tangent kernel by

$$K_\theta(\epsilon, \epsilon') := J_\theta(\epsilon) J_\theta(\epsilon')^\top,$$

and its integral operator by

$$(\Theta_\theta v)(\epsilon) := \int_X K_\theta(\epsilon, \epsilon') v(\epsilon') p_{\text{prior}}(\epsilon').$$

¹⁸From the **Riesz representation theorem** from functional analysis, we know that every drift field considered as a member of H^* has one corresponding element from H such that the inner product defined on H can mimic the operation of that functional, insofar as $\mathbf{V} \in H^*$ has a corresponding $f_{\mathbf{V}} \in H$ such that $\mathbf{V}(x) = \langle x, f_{\mathbf{V}} \rangle \forall x \in H$. This perfect corresponds to the definition of mobility tensors, that they are operators from the cotangent space to tangent space: $M(p) : T_p^* M \longrightarrow T_p M$

For convenience, define the bounded linear operator

$$A_\theta : \mathbb{R}^P \longrightarrow \mathcal{H}, \quad (A_\theta a)(\epsilon) := J_\theta(\epsilon)a.$$

Its Hilbert adjoint is

$$A_\theta^* v = \int_X J_\theta(\epsilon)^\top v(\epsilon) p_{\text{prior}}(\epsilon),$$

where the integral is well-defined by the Cauchy–Schwarz inequality. Consequently,

$$\Theta_\theta = A_\theta A_\theta^*.$$

Let

$$\mathcal{V}_\theta := \text{ran } A_\theta$$

denote the space of realizable latent velocity fields.

Lemma 7 (The NTK integral operator is self-adjoint). *For every $v, w \in \mathcal{H}$,*

$$\langle v, \Theta_\theta w \rangle_{\mathcal{H}} = \langle \Theta_\theta v, w \rangle_{\mathcal{H}}.$$

Proof. Using the factorization $\Theta_\theta = A_\theta A_\theta^*$ and the defining property of the Hilbert adjoint, we obtain

$$\begin{aligned} \langle v, \Theta_\theta w \rangle_{\mathcal{H}} &= \langle v, A_\theta A_\theta^* w \rangle_{\mathcal{H}} \\ &= \langle A_\theta^* v, A_\theta^* w \rangle_{\mathbb{R}^P} \\ &= \langle A_\theta A_\theta^* v, w \rangle_{\mathcal{H}} \\ &= \langle \Theta_\theta v, w \rangle_{\mathcal{H}}. \end{aligned}$$

Hence Θ_θ is self-adjoint.

At the kernel level, the corresponding symmetry is

$$K_\theta(\epsilon, \epsilon') = K_\theta(\epsilon', \epsilon)^\top,$$

since

$$(J_\theta(\epsilon') J_\theta(\epsilon)^\top)^\top = J_\theta(\epsilon) J_\theta(\epsilon')^\top.$$

□

Lemma 8 (The NTK integral operator is linear). *For every $v, w \in \mathcal{H}$ and $\lambda, \mu \in \mathbb{R}$,*

$$\Theta_\theta(\lambda v + \mu w) = \lambda \Theta_\theta v + \mu \Theta_\theta w.$$

Moreover, the form

$$b_\theta(v, w) := \langle v, \Theta_\theta w \rangle_{\mathcal{H}}$$

is a symmetric bilinear form on $\mathcal{H}$.

Proof. By linearity of the integral, for p_{prior} -almost every ϵ ,

$$\begin{aligned} (\Theta_\theta(\lambda v + \mu w))(\epsilon) &= \int_X K_\theta(\epsilon, \epsilon') (\lambda v(\epsilon') + \mu w(\epsilon')) p_{\text{prior}}(\epsilon') \\ &= \lambda \int_X K_\theta(\epsilon, \epsilon') v(\epsilon') p_{\text{prior}}(\epsilon') \\ &\quad + \mu \int_X K_\theta(\epsilon, \epsilon') w(\epsilon') p_{\text{prior}}(\epsilon') \\ &= \lambda (\Theta_\theta v)(\epsilon) + \mu (\Theta_\theta w)(\epsilon). \end{aligned}$$

Thus Θ_θ is linear.

Bilinearity of b_θ follows from linearity of Θ_θ and bilinearity of the real Hilbert inner product. Its symmetry follows from self-adjointness:

$$b_\theta(v, w) = \langle v, \Theta_\theta w \rangle_{\mathcal{H}} = \langle \Theta_\theta v, w \rangle_{\mathcal{H}} = b_\theta(w, v).$$

□

Lemma 9 (Positivity of the NTK integral operator). *The operator Θ_θ is positive semidefinite on $\mathcal{H}$. More precisely, for every $v \in \mathcal{H}$,*

$$\langle v, \Theta_\theta v \rangle_{\mathcal{H}} = \|A_\theta^* v\|_{\mathbb{R}^P}^2 \geq 0.$$

Furthermore,

$$\ker \Theta_\theta = \mathcal{V}_\theta^\perp, \quad \text{ran } \Theta_\theta = \mathcal{V}_\theta,$$

and the restriction on the neural realizable velocities

$$\Theta_\theta|_{\mathcal{V}_\theta} : \mathcal{V}_\theta \longrightarrow \mathcal{V}_\theta$$

is positive definite.

Proof. For every $v \in \mathcal{H}$, the factorization $\Theta_\theta = A_\theta A_\theta^*$ gives

$$\begin{aligned} \langle v, \Theta_\theta v \rangle_{\mathcal{H}} &= \langle v, A_\theta A_\theta^* v \rangle_{\mathcal{H}} \\ &= \langle A_\theta^* v, A_\theta^* v \rangle_{\mathbb{R}^P} \\ &= \|A_\theta^* v\|_{\mathbb{R}^P}^2 \geq 0. \end{aligned}$$

Equivalently,

$$\langle v, \Theta_\theta v \rangle_{\mathcal{H}} = \sum_{i=1}^P \left(\int_X \partial_{\theta_i} f(\epsilon) \cdot v(\epsilon) p_{\text{prior}}(d\epsilon) \right)^2.$$

This proves positive semidefiniteness.

The same identity shows that

$$\Theta_\theta v = 0 \iff A_\theta^* v = 0.$$

Indeed, the forward implication follows by taking the inner product with v , and the reverse implication follows from the factorization.

By the defining property of the adjoint,

$$A_\theta^* v = 0 \iff \langle A_\theta a, v \rangle_{\mathcal{H}} = 0 \text{ for every } a \in \mathbb{R}^P.$$

Therefore,

$$\ker \Theta_\theta = \ker A_\theta^* = (\text{ran } A_\theta)^\perp = \mathcal{V}_\theta^\perp.$$

Since $\Theta_\theta = A_\theta A_\theta^*$, its range is contained in $\mathcal{V}_\theta$. Now let $v \in \mathcal{V}_\theta$ satisfy

$$\langle v, \Theta_\theta v \rangle_{\mathcal{H}} = 0.$$

The preceding identities imply that $v \in \mathcal{V}_\theta^\perp$. Hence

$$v \in \mathcal{V}_\theta \cap \mathcal{V}_\theta^\perp = \{0\}.$$

Thus the restriction of Θ_θ to $\mathcal{V}_\theta$ is positive definite and, in particular, injective.

Because $\mathcal{V}_\theta$ is finite-dimensional, this restriction is also surjective onto $\mathcal{V}_\theta$. Consequently,

$$\text{ran } \Theta_\theta = \mathcal{V}_\theta.$$

□

Remark 10. *If $\mathcal{H}$ is infinite-dimensional and P is finite, then Θ_θ cannot be positive definite on all of $\mathcal{H}$, since*

$$\text{rank } \Theta_\theta \leq P$$

and $\mathcal{V}_\theta^\perp$ contains nonzero elements. Strict positive definiteness holds on the realizable latent velocity space $\mathcal{V}_\theta$.

Lemma 11. *(The neural tangent kernel integral operator is bounded)*

Proof. See appendix 9.1 for a proof that the NTK integral operator is a **Hilbert-Schmidt** operator, whence follows that it is a bounded operator. □

Proof of the theorem. By the preceding lemmas, Θ_θ is bounded, self-adjoint, and positive semidefinite, with

$$\text{ran } \Theta_\theta = \mathcal{V}_\theta.$$

Hence $M_\theta = \Theta_\theta \mathcal{R}_\mathcal{H}^{-1}$ is a bounded linear map from $\mathcal{H}^*$ to $\mathcal{V}_\theta$.

Let $\alpha, \beta \in \mathcal{H}^*$, and write

$$v = \mathcal{R}_\mathcal{H}^{-1}\alpha, \quad w = \mathcal{R}_\mathcal{H}^{-1}\beta.$$

Then

$$\alpha(M_\theta\beta) = \langle v, \Theta_\theta w \rangle_\mathcal{H} = \langle w, \Theta_\theta v \rangle_\mathcal{H} = \beta(M_\theta\alpha),$$

and

$$\alpha(M_\theta\alpha) = \langle v, \Theta_\theta v \rangle_\mathcal{H} \geq 0.$$

Therefore M_θ is a symmetric, positive-semidefinite mobility.

Since $\mathcal{V}_\theta$ is finite-dimensional, it is a closed Hilbert subspace of $\mathcal{H}$. For every nonzero $\alpha \in \mathcal{V}_\theta^*$, set $v = \mathcal{R}_{\mathcal{V}_\theta}^{-1}\alpha$. Then $v \neq 0$, and

$$\alpha(M_\theta^\mathcal{V}\alpha) = \langle v, \Theta_\theta v \rangle_\mathcal{H} > 0.$$

Symmetry follows in the same way. Thus $M_\theta^\mathcal{V}$ is a symmetric, positive-definite mobility. $\square$

Now that we have proven that the NTK is a mobility tensor, one would ask, what relevance or implication does it have? The insight is in that Onsager geometry and Riemannian geometry are directly related, so this discovery is indispensable towards our goal of showing that a Riemannian metric exists on the graph.

In a quadratic Onsager structure, the connection between mobility and metric is expressed through a dissipation potential. Recall from physics that the power a force exerts is its dot product with the velocity of the object the force exerts on:

$$P = F \cdot v \tag{78}$$

On a space where the mobility is invertible, the driving force required to sustain a velocity v is given by

$$F = M^{-1}v \tag{79}$$

In overdamped force balance, the power supplied by the driving force equals the dissipated power:

$$P = F \cdot v = (M^{-1}v)v = v^\top (M^{-1})^\top v = v^\top M^{-1}v \tag{80}$$

(Note: $(M^{-1})^\top = M^{-1}$ since M is symmetric)

Note that this quadratic relation is exactly algebraically equivalent to how a Riemannian metric acts on one single vector. Conceptually, the inverse of the mobility tensor can be interpreted as the resistance tensor, which measures the opposite relation: given a desired velocity, how much force is required to realize it. The inverse mobility defines the metric's vector-to-covector map, and its associated quadratic form measures the dissipation rate in this Onsager model. In a quadratic Onsager structure, the metric measures the instantaneous dissipation cost of a prescribed velocity:

$$\Psi(v) = \frac{1}{2}g(v, v) = \frac{1}{2}v^\top M^{-1}v^\top, P_{\text{diss}} = g(v, v) \tag{81}$$

Thus, the most important conceptual bridge this discussion yields would be

$$\boxed{\text{Mobility}^{-1} = \text{Metric}} \tag{82}$$

So, applying this relation to

$$u = \Theta \mathbf{V}, \quad (83)$$

we would have

$$\mathbf{V} = \Theta^{-1} u \quad (84)$$

So, using the language from Onsager geometry, the instantaneous power of a neural architecture realizing the drift field with the actual induced velocity field would be given by the inverse restricted to the neural realizable velocity space:

$$\langle u, (\Theta_\theta|_{\mathcal{V}_\theta})^{-1} u \rangle_H = \min_{\dot{\theta}: u=J_\theta \dot{\theta}} \|\dot{\theta}\|^2 \quad (85)$$

The interpretation of this inner product then immediately follows: it is a measurement of the minimum squared parameter velocity required to produce u .

4.2.2 Deriving a Riemannian Metric on Parameter-Distribution Graph

The inverse of the NTK restricted to $\mathcal{V}_\theta$ defines an inner product on realizable latent velocities. Pulling this inner product back through the parameter Jacobian defines a Riemannian metric on the graph wherever that Jacobian is injective. The inverse of the neural tangent kernel induces a metric in the following way. Recall from Riemannian geometry that the Riemannian metric defines an inner product as follows, whence geometric concepts such as length and angles emerge, as:

$$g_p(u, u) = \langle u, M^{-1} u \rangle = u_i^\top M_{ij}^{-1} u_j \quad (86)$$

The analog to this quadratic form, as mentioned in the previous section, in this high-dimensional and smooth manifold would then be a double integral over the entire space, where every point is evaluated on that quadratic form given by the inverse of the NTK. But, here, a remark is due: the NTK as an integral operator is not invertible on all of $L^2(q_\theta; \mathbb{R}^d)$, since its rank is at most p . A finite rank operator can never be surjective onto an infinite-dimensional space. So here, we introduce an important mathematical tool to substitute the full inverse, which cannot exist everywhere: the **Moore-Penrose Pseudoinverse**, denoted by A^+ for a matrix A . The Moore-Penrose Pseudoinverse is the most widely known generalization of the inverse matrix that exists uniquely for all rectangular matrices, and thus enables us to easily invert the NTK in matrix form. In our case, this pseudoinverse proves to be a perfect tool, since here our NTK has closed range, so its pseudoinverse is bounded on the entire space of all neural realizable velocities.¹⁹

From the earlier identity that $u = \Theta \mathbf{V}$, we can recover the velocity field via the pseudo-inverse of the NTK as follows:

$$\mathbf{V} = (\Theta^+ u)(\epsilon) = \int_X \Theta_\theta^+(\epsilon, \epsilon') u(f_\theta(\epsilon')) dp_{\text{prior}}(\epsilon') \quad (87)$$

And then, the inner product applies this to the transpose of u :

$$\langle u, \mathbf{V} \rangle = \langle u, \Theta^+ u \rangle = \int_X u(f_\theta(\epsilon))^\top (\Theta_\theta^+ u)(\epsilon) dp_{\text{prior}}(\epsilon) \quad (88)$$

¹⁹A quick discussion here: if we set $\mathcal{V}_\theta = \text{Im}(\Theta_\theta) = \text{Im}(J_\theta f_\theta)$, which would be the set of all neural realizable velocity fields, all we need is an inversion of the NTK as an operator $\Theta_\theta^+ : \mathcal{V}_\theta \rightarrow \mathcal{V}_\theta$, instead of all of $L^2(q_\theta; \mathbb{R}^d)$. When restricted to $\mathcal{V}_\theta$, we have that $\Theta_\theta \Theta_\theta^+ = \Theta_\theta^+ \Theta_\theta = P_{(\ker \Theta_\theta)^\perp}$, which is the orthogonal projection onto the complement of the kernel of the NTK. Note that the kernel of the NTK would represent null directions that the neural architecture cannot realize, so its complement is simply all the realizable directions. So, when restricted to the realizable space, $P_{(\ker \Theta_\theta)^\perp} = I$.

Expanding,

$$= \int_X u(f_\theta(\epsilon))^\top \int_X \Theta_\theta^+(\epsilon, \epsilon') u(f_\theta(\epsilon')) dp_{\text{prior}}(\epsilon') dp_{\text{prior}}(\epsilon), \quad (89)$$

and then taking the inner integral outside,

$$\int_X \int_X u(f_\theta(\epsilon))^\top \Theta_\theta^+(\epsilon, \epsilon') u(f_\theta(\epsilon')) dp_{\text{prior}}(\epsilon) dp_{\text{prior}}(\epsilon') \quad (90)$$

This gives rise to a definition of an inner product on the space of all neural realizable velocity fields:

Definition 12 (Inverse-NTK-induced Inner product on velocity field space). *Using the Moore-Penrose pseudoinverse of the neural tangent kernel, the inner product on the space of neural realizable velocity fields $\mathcal{V}_\theta$, $\langle \cdot, \cdot \rangle : \mathcal{V}_\theta \times \mathcal{V}_\theta \rightarrow \mathbb{R}$, is hereby defined as*

$$\langle u_1, u_2 \rangle_{\Theta_\theta^+} = \langle u_1, \Theta_\theta^+ u_2 \rangle = \int_X \int_X u_1(f_\theta(\epsilon))^\top \Theta_\theta^+(\epsilon, \epsilon') u_2(f_\theta(\epsilon')) dp_{\text{prior}}(\epsilon) dp_{\text{prior}}(\epsilon')$$

Thus, as we can see, the inverse of the neural tangent kernel induces an inner product on the space of velocity fields. But our ultimate goal is not a geometry on the velocity field, but on the final distribution space. Instead of operating directly on the distribution space, we can take advantage of what we have constructed by first pulling the velocity field to something more manageable and then pushing it forward.

On the parameter space, then, we can define a metric on the space of parameter perturbations via the inner product we defined earlier on the velocity field space

$$\tilde{g}_\theta(\dot{\theta}_1, \dot{\theta}_2) := \langle u_1, u_2 \rangle_{\Theta_\theta^+}, \text{ where } u_i = J_\theta f_\theta \dot{\theta}_i \quad (91)$$

Now, we can let the geometry on the distribution space descend onto the parameter space via the mapping $\Phi : \mathbb{R}_{\text{reg}}^p \rightarrow M_f^{p_{\text{prior}}}$ via the pushforward operator we saw earlier: $\Phi = (f_{(\cdot)})_{\#} p_{\text{prior}}$, whose differential pushforward is established to be injective since the domain is on the regular parameter space. This is where we invoke the graph construction since we must know which parameter is behind the corresponding distributional deformation:

$$g_{q_\theta}(\dot{q}_1, \dot{q}_2) := \tilde{g}_\theta((d\Phi)^{-1}(\dot{q}_1), (d\Phi)^{-1}(\dot{q}_2)) = \langle J_\theta f_\theta(d\Phi)^{-1}(\dot{q}_1), J_\theta f_\theta(d\Phi)^{-1}(\dot{q}_2) \rangle_{\Theta_\theta^+} \quad (92)$$

So, we are defining a metric on the parameter-distribution graph by pulling the distributional deformations back to the parameter perturbation that gave rise to it, and then pushing it towards the velocity space inner product. We can formally write this as a double integral on the velocity field as:

Definition 13 (Pseudoinverse of NTK Pullback Induced Riemannian Metric on the parameter-distribution graph manifold). *We hereon refer to the following quantity, which defines a bilinear map on the tangent space of the parameter-distribution graph manifold,*

$$g_q(\dot{q}_1, \dot{q}_2) : T_q \Gamma_\Phi \times T_q \Gamma_\Phi \rightarrow \mathbb{R}$$

as the **NTK-induced metric**:

$$g_q(\dot{q}_1, \dot{q}_2) = \int_X \int_X \underbrace{J_\theta f_\theta(d\Phi)^{-1}(\dot{q}_1)(f_\theta(\epsilon))^\top}_{u_1(f_\theta(\epsilon))^\top} \Theta_\theta^+(\epsilon, \epsilon') \underbrace{J_\theta f_\theta(d\Phi)^{-1}(\dot{q}_2)(f_\theta(\epsilon'))}_{u_2(f_\theta(\epsilon'))} dp_{\text{pr}}(\epsilon) dp_{\text{pr}}(\epsilon')$$

Remark There is a very beautiful fact related to this metric: that by a simple algebraic manipulation, it reduces back to standard Euclidean metric on the $\mathbb{R}^p$ space: $\langle \dot{\theta}_1, \dot{\theta}_2 \rangle = \dot{\theta}_1^\top \dot{\theta}_2$. The significance of this is twofold: first, conceptually, it means that this metric is at its base both correct and simple, since it is no different than a simple Euclidean metric on the Euclidean parameter space. This reflects the fact that this metric, though requiring the NTK to represent at the distribution level, really is fundamentally measuring nothing else other than the Euclidean metric on the underlying parameter perturbations giving rise to motion at the distribution level. Additionally, it also would make proving the required properties for a bilinear form to be a Riemannian metric, i.e., bilinearity, symmetry, positive definiteness, and smoothness, effortless, since the Euclidean norm is long known to support all of those. So, let us first show this reduction.

Theorem 14 (The NTK-induced metric is equivalent to the Euclidean metric on the parameter space $\mathbb{R}^p$). *That is, $g_q(\dot{q}_1, \dot{q}_2) = \langle \dot{\theta}_1, \dot{\theta}_2 \rangle_{\mathbb{R}^p} = \dot{\theta}_1^\top \dot{\theta}_2$*

Proof. Once again recall from the discussion in section 4.2.1 that the space of velocity fields are Hilbert spaces. So let

$$\mathcal{H}_\theta := L^2(p_{\text{prior}}; \mathbb{R}^d)$$

denote the Hilbert space of velocity fields, and define the parameter-Jacobian operator

$$J_\theta : T_\theta \mathbb{R}^p \longrightarrow \mathcal{H}_\theta, \quad J_\theta \dot{\theta} = \left[\epsilon \longmapsto J_\theta f_\theta(\epsilon) \dot{\theta} \right].$$

Let

$$J_\theta^* : \mathcal{H}_\theta \longrightarrow T_\theta \mathbb{R}^p$$

denote its Hilbert-space adjoint. The NTK integral operator can then be written as

$$\Theta_\theta = J_\theta J_\theta^*.$$

Let

$$\Phi : \Theta_{\text{reg}} \longrightarrow \mathcal{P}_{q_0}^s(\mathbb{R}^d)$$

denote the parameter-to-distribution realization map. Consider distributional tangent vectors

$$\dot{q}_i = d\Phi_\theta[\dot{\theta}_i], \quad i \in \{1, 2\},$$

where

$$\dot{\theta}_i = (d\Phi_\theta)^{-1}[\dot{q}_i]$$

on the regular parameter set. The corresponding architecture-realizable velocity fields are

$$u_i = J_\theta \dot{\theta}_i.$$

The NTK-induced metric is defined by

$$g_q(\dot{q}_1, \dot{q}_2) = \left\langle u_1, \Theta_\theta^\dagger u_2 \right\rangle_{\mathcal{H}_\theta}.$$

Substituting

$$u_i = J_\theta \dot{\theta}_i$$

and

$$\Theta_\theta = J_\theta J_\theta^*$$

gives

$$g_q(\dot{q}_1, \dot{q}_2) = \left\langle J_\theta \dot{\theta}_1, (J_\theta J_\theta^*)^\dagger J_\theta \dot{\theta}_2 \right\rangle_{\mathcal{H}_\theta}.$$

Using the defining property of the adjoint,

$$\langle J_\theta a, v \rangle_{\mathcal{H}_\theta} = \langle a, J_\theta^* v \rangle_{\mathbb{R}^p},$$

we obtain

$$g_{\mathbf{q}}(\dot{\mathbf{q}}_1, \dot{\mathbf{q}}_2) = \left\langle \dot{\theta}_1, J_{\theta}^* (J_{\theta} J_{\theta}^*)^{\dagger} J_{\theta} \dot{\theta}_2 \right\rangle_{\mathbb{R}^p}.$$

The Moore–Penrose pseudoinverse satisfies the operator identity

$$J_{\theta}^* (J_{\theta} J_{\theta}^*)^{\dagger} J_{\theta} = P_{(\ker J_{\theta})^{\perp}},$$

where

$$P_{(\ker J_{\theta})^{\perp}} : \mathbb{R}^p \longrightarrow (\ker J_{\theta})^{\perp}$$

is the orthogonal projection onto the complement of the kernel of J_{θ} . Therefore,

$$g_{\mathbf{q}}(\dot{\mathbf{q}}_1, \dot{\mathbf{q}}_2) = \left\langle \dot{\theta}_1, P_{(\ker J_{\theta})^{\perp}} \dot{\theta}_2 \right\rangle_{\mathbb{R}^p}.$$

On the regular parameter set, $d\Phi_{\theta}$ is injective. Since

$$J_{\theta} \dot{\theta} = 0 \implies d\Phi_{\theta}[\dot{\theta}] = -\nabla \cdot (q_{\theta} J_{\theta} \dot{\theta}) = 0,$$

we have

$$\ker J_{\theta} \subseteq \ker d\Phi_{\theta} = \{0\}.$$

Consequently,

$$\ker J_{\theta} = \{0\},$$

and hence

$$P_{(\ker J_{\theta})^{\perp}} = I_{\mathbb{R}^p}.$$

It follows that

$$g_{\mathbf{q}}(\dot{\mathbf{q}}_1, \dot{\mathbf{q}}_2) = \left\langle \dot{\theta}_1, \dot{\theta}_2 \right\rangle_{\mathbb{R}^p} = \dot{\theta}_1^{\top} \dot{\theta}_2.$$

□

Thus, on the regular parameter set, the NTK-pseudoinverse metric, defined on the neural realizable deformations, is equivalent to the Euclidean parameter metric expressed through the architecture-realizable velocity and distributional coordinates.

Since we have shown how this metric reduces to the Euclidean metric on the Euclidean regular parameter space, it inherits all the properties of it being a Riemannian metric. Thus, we have now proven that $(\Gamma_{\Phi}, g_{\mathbf{q}})$ is a smooth manifold diffeomorphic to the regular parameter space equipped with a Riemannian metric, and thus is a Riemannian manifold. This concludes our theoretical construction of the geometric framework.

Let us briefly discuss one significance of this work in light of Deng’s advancement. Drift-based training as proposed by Deng et al. [2026] makes distributional evolution visible while leaving its realization-dependent parameter effort geometrically uncharacterized. This graph manifold of realizable distributions equipped with a metric directly allows for computation of neural realization effort via the usual length definition arising

from a Riemannian metric: $\int_0^T \sqrt{g_{\mathbf{q}(t)}(\dot{\mathbf{q}}(t), \dot{\mathbf{q}}(t))} dt$, opening a concrete method of calculating neural architecture realization effort for a distribution evolution $q_{\theta(t)}, t \in [0, T]$.

5 Distributional Inclusion of Convolutional Neural Networks in Transformers

Now that we have established the geometric basis for distribution evolution for a neural architecture, let us proceed to show one example of an important application of this framework: showing that the graph manifold of realizable distributions of one architecture is the submanifold of another, thereby revealing that the distributions that the previous architecture can realize can be realized by the latter. The following, then, can be taken as an example of an important application of the geometric framework we have built.

Let $\Omega = [H] \times [W]$ be a finite grid of a CNN. For a tensor $X \in \mathbb{R}^{H \times W \times D_{\text{in}}}$, we write $X_q \in \mathbb{R}^{D_{\text{in}}}$ for the feature vector at position $q \in \Omega$. A convolutional layer with offset set $\Delta \subset \mathbb{Z}^2$ has the form

$$\text{Conv}(X)_q = \sum_{\delta \in \Delta} X_{q+\delta} W_\delta + b, \quad (93)$$

where $W_\delta \in \mathbb{R}^{D_{\text{in}} \times D_{\text{out}}}$ and $b \in \mathbb{R}^{D_{\text{out}}}$.

Now consider a single MHSA layer evaluated at the same query position $q \in \Omega$. For head $h \in [N_h]$, let $a_{q,k}^{(h)}$ denote the attention weight assigned from this query position q to a key position $k \in \Omega$. The head first aggregates the input features as $\sum_{k \in \Omega} a_{q,k}^{(h)} X_k$, and then applies its effective channel-mixing matrix

$$M_h := W_{\text{val}}^{(h)} W_{\text{out}}^{(h)} \in \mathbb{R}^{D_{\text{in}} \times D_{\text{out}}}. \quad (94)$$

Thus the MHSA output at the same position q can be written as

$$\text{MHSA}(X)_q = b_{\text{out}} + \sum_{h=1}^{N_h} \left(\sum_{k \in \Omega} a_{q,k}^{(h)} X_k \right) M_h. \quad (95)$$

The following argument builds on Cordonnier et al. [2019] that relative-position multi-head self-attention can realize convolutional layers when equipped with sufficiently many heads.

Lemma 15 (Layer-wise convolution-to-attention reparameterization). *Let Conv be a convolutional layer with offset set Δ . Suppose an MHSA layer satisfies the following two conditions. First, it has at least one head per convolutional offset: $N_h \geq |\Delta|$. For each $\delta \in \Delta$, choose one head h_δ . Second, each chosen head performs attention:*

$$a_{q,k}^{(h_\delta)} = \mathbf{1}\{k = q + \delta\}. \quad (96)$$

Assume also that the head dimension is large enough so that each effective head matrix M_{h_δ} can realize any matrix in $\mathbb{R}^{D_{\text{in}} \times D_{\text{out}}}$; for instance, it suffices to assume $D_{h_\delta} \geq D_{\text{out}}$. Then the MHSA layer can be parameterized so that $\text{MHSA}(X) = \text{Conv}(X)$ for every input X .

Proof. We construct the MHSA parameters directly from the convolutional parameters. For each offset $\delta \in \Delta$, assign one head h_δ . By hard relative attention, this head extracts exactly the feature vector at relative offset δ :

$$\sum_{k \in \Omega} a_{q,k}^{(h_\delta)} X_k = X_{q+\delta}.$$

Now set the effective matrix of this head equal to the convolutional filter at the same offset:

$$M_{h_\delta} = W_\delta.$$

This is possible under $D_{h_\delta} \geq D_{\text{out}}$; for example, one may take

$$W_{\text{val}}^{(h_\delta)} = [W_\delta \quad 0], \quad W_{\text{out}}^{(h_\delta)} = \begin{bmatrix} I_{D_{\text{out}}} \\ 0 \end{bmatrix}.$$

Then $W_{\text{val}}^{(h_\delta)} W_{\text{out}}^{(h_\delta)} = W_\delta$. Set $b_{\text{out}} = b$ and set unused heads, if any, to have zero effective matrix. Therefore

$$\text{MHSA}(X)_q = \sum_{\delta \in \Delta} X_{q+\delta} W_\delta + b = \text{Conv}(X)_q.$$

Since this holds for every position q and every input tensor X , the two layers are identical as functions. $\square$

Theorem 16 (CNN-to-Transformer parameter transformation). *Fix a CNN generator architecture whose ℓ -th convolutional layer has offset set Δ_ℓ and output channel dimension D_ℓ^{out} . Construct a hard relative-attention Transformer architecture by replacing each convolutional layer with an MHSA layer satisfying*

$$N_\ell \geq |\Delta_\ell|, \quad D_{h,\ell} \geq D_\ell^{\text{out}}, \quad (97)$$

and by assigning one attention head to each offset $\delta \in \Delta_\ell$. Assume that nonlinear activations are shared between the two architectures. Then there exists an explicit parameter map

$$S : \mathbb{R}_{\text{CNN}}^p \rightarrow \mathbb{R}_{\text{Tr}}^p \quad (98)$$

such that for every CNN parameter θ ,

$$f_{S(\theta)}^{\text{Tr}}(z) = f_\theta^{\text{CNN}}(z) \quad \text{for all } z, \quad (99)$$

where CNN is the convolutional generator and Tr is the Transformer generator. Consequently, the CNN generator function class is contained in the Transformer generator function class:

$$\mathcal{G}_{\text{CNN}} \subseteq \mathcal{G}_{\text{Tr}}. \quad (100)$$

Proof. Fix an arbitrary CNN parameter θ . We build the Transformer parameter $S(\theta)$ layer by layer. In the ℓ -th CNN layer, write the convolutional part as

$$C_\ell(X)_q = \sum_{\delta \in \Delta_\ell} X_{q+\delta} W_{\ell,\delta} + b_\ell.$$

By Lemma 15, there is an MHSA layer A_ℓ such that $A_\ell(X) = C_\ell(X)$ for every X . Let the CNN block be $T_\ell = \sigma_\ell \circ C_\ell$, where σ_ℓ denotes the nonlinear activations. Define the corresponding Transformer block by $\tilde{T}_\ell = \sigma_\ell \circ A_\ell$. Since $A_\ell = C_\ell$ as functions, we have $\tilde{T}_\ell = T_\ell$ as functions.

The full CNN generator is a composition $f_\theta^{\text{CNN}} = T_L \circ \dots \circ T_1$, while the constructed Transformer generator is $f_{S(\theta)}^{\text{Tr}} = \tilde{T}_L \circ \dots \circ \tilde{T}_1$. Because each block agrees with its CNN counterpart, their compositions agree:

$$f_{S(\theta)}^{\text{Tr}}(z) = f_\theta^{\text{CNN}}(z) \quad \text{for all } z.$$

Since θ was arbitrary, every CNN generator in the considered class is represented by some Transformer generator. Hence $\mathcal{G}_{\text{CNN}} \subseteq \mathcal{G}_{\text{Tr}}$. $\square$

Lemma 17 (Equality of maps implies equality of pushforwards). *Let $f, g : \mathcal{Z} \rightarrow \mathcal{X}$ be measurable maps and let p_{prior} be a probability measure on $\mathcal{Z}$. If $f(z) = g(z)$ for every $z \in \mathcal{Z}$, then $f_{\#}p_{\text{prior}} = g_{\#}p_{\text{prior}}$.*

Proof. Let $A \subseteq \mathcal{X}$ be any measurable set. By definition of pushforward,

$$(f_{\#}p_{\text{prior}})(A) = p_{\text{prior}}(f^{-1}(A)), \quad (g_{\#}p_{\text{prior}})(A) = p_{\text{prior}}(g^{-1}(A)).$$

Since $f(z) = g(z)$ for every z , we have $f^{-1}(A) = g^{-1}(A)$. Therefore $(f_{\#}p_{\text{prior}})(A) = (g_{\#}p_{\text{prior}})(A)$ for every measurable A , and hence $f_{\#}p_{\text{prior}} = g_{\#}p_{\text{prior}}$. $\square$

Theorem 18 (CNN distribution locus is set included in the Transformer distribution locus). *Using our definition of probability distribution locus in section (5):*

$$\mathfrak{M}_{\text{CNN}}^{p_{\text{prior}}} := \{(f_\theta^{\text{CNN}})_{\#}p_{\text{prior}} : \theta \in \mathbb{R}_{\text{reg}}^{\text{CNN}}\}, \quad \mathfrak{M}_{\text{Tr}}^{p_{\text{prior}}} := \{(f_\phi^{\text{Tr}})_{\#}p_{\text{prior}} : \phi \in \mathbb{R}_{\text{reg}}^{\text{Tr}}\}. \quad (101)$$

Under the assumptions of Theorem 16,

$$\mathfrak{M}_{\text{CNN}}^p \subseteq \mathfrak{M}_{\text{Tr}}^p. \quad (102)$$

Proof. Take any $q \in \mathfrak{M}_{\text{CNN}}^p$. Then $q = (f_\theta^{\text{CNN}})_{\#}p_{\text{prior}}$ for some $\theta \in \Theta_{\text{CNN}}$. By Theorem 16, there exists $\phi = S(\theta)$ such that $f_\phi^{\text{Tr}} = f_\theta^{\text{CNN}}$ as functions. By Lemma 17,

$$(f_\phi^{\text{Tr}})_{\#}p_{\text{prior}} = (f_\theta^{\text{CNN}})_{\#}p_{\text{prior}} = q_\theta.$$

Thus $q_\theta \in \mathfrak{M}_{\text{Tr}}^{p_{\text{prior}}}$. Since q_θ was arbitrary, $\mathfrak{M}_{\text{CNN}}^{p_{\text{prior}}} \subseteq \mathfrak{M}_{\text{Tr}}^{p_{\text{prior}}}$. $\square$

Corollary 19 (CNN distributional curves are Transformer-realizable). *Let $t \mapsto \theta(t)$ be any CNN training trajectory. Under the assumptions above, there exists a Transformer trajectory $t \mapsto \phi(t)$ such that*

$$(f_{\phi(t)}^{\text{Tr}})_{\#}p_{\text{prior}} = (f_{\theta(t)}^{\text{CNN}})_{\#}p_{\text{prior}} \quad \forall t. \quad (103)$$

Proof. Set $\phi(t) := S(\theta(t))$. By Theorem 16, $f_{\phi(t)}^{\text{Tr}} = f_{\theta(t)}^{\text{CNN}}$ for every t . Taking pushforwards by ρ gives the claim. $\square$

But so far we have only shown that the relationship between the two distribution loci is that of a set inclusion. We have not yet utilized the geometry on the two manifolds, which is rather the central result of our paper. We want to further show that Γ_{CNN} is an immersion into Γ_{Tr} . To prove this, we can capitalize upon the results from corollary 19 that $\phi(t)$ is given by the function composition between S and $\theta(t)$.

Theorem 20 (Parameter-Distribution Graph Manifolds of CNN realizable distributions are immersed submanifolds into the Parameter-Distribution Graph Manifolds of Transformer realizable distributions).

$$\Gamma_{\text{CNN}} \hookrightarrow \Gamma_{\text{Tr}}$$

Lemma 21. (*The map embedding S is an immersion*)

Proof. Smoothness follows from the explicit layerwise parameter construction in Theorem 10. It remains to show that

$$dS_{\theta} : T_{\theta} \mathbb{R}_{\text{reg}}^{\text{CNN}} \longrightarrow T_{S(\theta)} \mathbb{R}_{\text{reg}}^{\text{Tr}}$$

is injective for every $\theta \in \Theta_C^{\text{reg}}$.

The identity

$$\Phi_T \circ S = \Phi_C$$

implies, by the chain rule,

$$d\Phi_T \circ dS = d\Phi_C|_{\theta}.$$

Suppose that $\dot{\theta} \in T_{\theta} \Theta_C^{\text{reg}}$ satisfies

$$dS_{\theta}[\dot{\theta}] = 0.$$

Then

$$d\Phi_T|_{S(\theta)}[0]$$

Because $\theta \in \Theta_C^{\text{reg}}$, the differential $d\Phi_C|_{\theta}$ is injective. Consequently,

$$\dot{\theta} = 0.$$

Therefore,

$$\ker(dS_{\theta}) = 0,$$

so dS_{θ} is injective at every $\theta \in \Theta_C^{\text{reg}}$. Hence S is a smooth immersion. $\square$

Lemma 22. (*The induced map between graph manifolds*).

Define

$$\tilde{S} : \Gamma_{\Phi_C} \longrightarrow \Gamma_{\Phi_T}$$

by

$$\tilde{S}(\theta, \Phi_C(\theta)) := (S(\theta), \Phi_T(S(\theta))).$$

Then $\tilde{S}$ is well-defined and smooth. Its differential is given by

$$d\tilde{S}_{(\theta, \Phi_C(\theta))} \left(\dot{\theta}, d\Phi_C|_{\theta}[\dot{\theta}] \right) = \left(dS_{\theta}[\dot{\theta}], d\Phi_T|_{S(\theta)}[dS_{\theta}[\dot{\theta}]] \right).$$

Proof. Because

$$S(\theta) \in \mathbb{R}_{\text{reg}}^{\text{Tr}},$$

the point

$$(S(\theta), \Phi_T(S(\theta)))$$

belongs to Γ_{Φ_T} . Thus $\tilde{S}$ is well-defined.

Let

$$\iota_C : \mathbb{R}_{\text{reg}}^{\text{Cnn}} \longrightarrow \Gamma_{\Phi_C}, \quad \iota_C(\theta) = (\theta, \Phi_C(\theta)),$$

and

$$\iota_T : \mathbb{R}_{\text{reg}}^{\text{Tr}} \longrightarrow \Gamma_{\Phi_T}, \quad \iota_T(\eta) = (\eta, \Phi_T(\eta))$$

denote the respective graph embeddings. By Proposition 5, these maps are diffeomorphisms onto their graph manifolds. The induced graph map can therefore be written as

$$\tilde{S} = \iota_T \circ S \circ \iota_C^{-1}.$$

Since ι_C^{-1} , S , and ι_T are smooth, $\tilde{S}$ is smooth.

Applying the chain rule to a graph tangent vector gives

$$d\tilde{S}_{(\theta, \Phi_C(\theta))} \left(\dot{\theta}, d\Phi_C|_{\theta}[\dot{\theta}] \right) = \left(dS_{\theta}[\dot{\theta}], d\Phi_T|_{S(\theta)}[dS_{\theta}[\dot{\theta}]] \right).$$

Using

$$d\Phi_T|_{S(\theta)} \circ dS_{\theta} = d\Phi_C|_{\theta},$$

this may equivalently be written as

$$d\tilde{S}_{(\theta, \Phi_C(\theta))} \left(\dot{\theta}, d\Phi_C|_{\theta}[\dot{\theta}] \right) = \left(dS_{\theta}[\dot{\theta}], d\Phi_C|_{\theta}[\dot{\theta}] \right).$$

Thus, the corresponding CNN and Transformer graph tangent vectors generally have different parameter components but the same distributional component. $\square$

Lemma 23. *(The induced graph map is an immersion).*

The induced map

$$\tilde{S} : \Gamma_{\Phi_C} \longrightarrow \Gamma_{\Phi_T}$$

is a smooth immersion.

Proof. By Lemma 16, $\tilde{S}$ is smooth. Let

$$\mathbf{v}_C = \left(\dot{\theta}, d\Phi_C|_{\theta}[\dot{\theta}] \right) \in T_{(\theta, \Phi_C(\theta))}\Gamma_{\Phi_C},$$

and suppose that

$$d\tilde{S}_{(\theta, \Phi_C(\theta))}[\mathbf{v}_C] = 0.$$

By Lemma 16,

$$d\tilde{S}_{(\theta, \Phi_C(\theta))}[\mathbf{v}_C] = \left(dS_{\theta}[\dot{\theta}], d\Phi_C|_{\theta}[\dot{\theta}] \right).$$

Therefore,

$$dS_{\theta}[\dot{\theta}] = 0.$$

Since S is an immersion by Lemma 15, dS_{θ} is injective, and hence

$$\dot{\theta} = 0.$$

It follows that

$$d\Phi_C|_{\theta}[\dot{\theta}] = 0$$

as well, so $\mathbf{v}_C = 0$. Thus,

$$\ker \left(d\tilde{S}_{(\theta, \Phi_C(\theta))} \right) = \{0\}.$$

Therefore, $d\tilde{S}$ is injective at every point, and $\tilde{S}$ is a smooth immersion. This proves Theorem 14. $\square$

This is a direct application of the graph manifold of neural architecture realizable distributions we have constructed in this paper. This result demonstrates the role of the parameter–distribution graph in architecture comparison. Although the global distributional loci need not be manifolds, the corresponding graph manifolds admit a well-defined immersion induced by the parameter transformation S . We hope this example we walked through can inspire manifold inclusion as a further methodology on comparing the relative distributional expressive power of different neural architectures. This is also a concrete example that demonstrates the value of the geometric framework we constructed, since geometric concepts such as submanifolds and immersions are borrowed over and combined with deep learning architectures.

6 Experimental Results

For our experiment, we intend to numerically approximate the NTK-induced metric in a discrete computational method exploiting the methods of finite differences. Then, we wish to verify a central claim in the theory portion of this paper: *is the NTK-induced metric at the distribution / velocity space level equal to the Euclidean metric at the parameter level?* This is both a theoretically important question and a question that would naturally invite empirical confirmation. If this hypothesis can be empirically verified, it will greatly lend credence to our construction of the NTK-metric, that the NTK does correctly represent distributional level deformation when pushed forward from the parameter level.

To perform this experiment, we intend to simulate the training of a neural network guided by drift fields, as proposed by Deng et al. We use SGD with the loss function defined by the drift field, which we again state here:

$$\mathcal{L} = \mathbb{E}_{\epsilon} [\|f_{\theta}(\epsilon) - \text{stopgrad}(f_{\theta}(\epsilon) + \mathbf{V}_{p,q\theta}(f_{\theta}(\epsilon))\|^2] \quad (104)$$

But, the next immediate question will be, how does one numerically approximate the drift field, which is present in this quantity. To do this, we must take a certain number of samples to approximate the two distributions p and q . p refers to the final distribution, so in our simulation, we would define a shape for the final distribution with a finite number of points, and then uniformly sample points, one by one, from that shape. For our experiment, we set the final, intended distribution to be the Swiss roll at the output dimension. For q , it is simply using the current neural network as the generator from the prior distribution, which we choose to be the Gaussian distribution on the output dimension, since it is easy to sample from. So, expanding out the definition of the drift field from Deng et al. [2026]’s paper,

$$\mathbf{V}_{p,q}(\mathbf{x}) = \frac{1}{Z_p Z_q} \mathbb{E}_{p,q} [k(\mathbf{x}, \mathbf{y}^+) k(\mathbf{x}, \mathbf{y}^-) (\mathbf{y}^+ - \mathbf{y}^-)] , \quad (105)$$

where $k(\cdot, \cdot) : \mathbb{R}^d \times \mathbb{R}^d \rightarrow \mathbb{R}$ is a kernel function that measures the similarity between the two samples. The kernel function used in Deng et al. [2026]’s paper is the Gaussian kernel,

$$k(\mathbf{x}, \mathbf{y}) = \exp \left(-\frac{1}{\tau} \|\mathbf{x} - \mathbf{y}\|_2 \right) \quad (106)$$

which we also adopt. We set τ , the temperature, to be 1.

Then, for each iteration, we would approximate the drift field with n samples via Monte Carlo:

$$\mathbf{V}_{p,q}(f_t(\epsilon)) \approx \frac{1}{Z_p Z_q n} \sum_{i=1}^n k(f_t(\epsilon), p_i) k(f_t(\epsilon), f_t(\epsilon_i)) (p_i - f_t(\epsilon_i)), \text{ where } \epsilon_i \sim \mathcal{N}(0, \mathbf{I}_{\text{in} \times \text{in}}), p_i \sim p \quad (107)$$

The normalization constants will also be computed along the way:

$$Z_p = \mathbb{E}_p[k(\mathbf{x}, \mathbf{y}^+)] \approx \frac{1}{n} \sum_{i=1}^n k(\mathbf{x}, p_i), \text{ where } p_i \sim p \quad (108)$$

$$Z_q = \mathbb{E}_q[k(\mathbf{x}, \mathbf{y}^-)] \approx \frac{1}{n} \sum_{i=1}^n k(\mathbf{x}, f(\epsilon_i)), \text{ where } \epsilon_i \sim \mathcal{N}(0, I_{\text{in} \times \text{in}}) \quad (109)$$

This gives us a Monte Carlo approximation of the drift field, which we then will use to compute the loss function that drives the training via SGD. In the meantime, to verify our hypothesis, we also would want to compute the NTK-metric at the velocity space for each iteration. To do this, let us discretize the NTK-metric by first noting that a velocity field, $u_t(f_{\theta_t}(\epsilon_i))$, is formally defined as $\lim_{\Delta t \rightarrow 0} \frac{f_{t+\Delta t}(\epsilon_i) - f_t(\epsilon_i)}{\Delta t}$. So, by the method of finite differences, in computational realization of training, we can only set $\Delta t = 1$. Then, we would have $u_t(f_{\theta}(\epsilon_i)) \approx f_{t+1}(\epsilon_i) - f_t(\epsilon_i)$. So, the velocity field in a computational setting is simply the difference between the neural network output in the next iteration from the previous. The numerical approximation for the inverse-NTK in a discrete setting is more deliberate. To discretize the NTK as a continuous integral operator, we will need to use the **Gram Matrix** representation of the NTK over the samples we have taken. We use the same n samples from earlier, and for each pair, we compute the outer product NTK for that pair. This will be an $nd \times nd$ symmetric matrix, since each entry is a matrix of dimension $d \times d$, with n rows and columns:

$$\mathbf{K}_t = \begin{bmatrix} \Theta_{\theta_t}(\epsilon_1, \epsilon_1) & \Theta_{\theta_t}(\epsilon_1, \epsilon_2) & \cdots & \Theta_{\theta_t}(\epsilon_1, \epsilon_n) \\ \Theta_{\theta_t}(\epsilon_2, \epsilon_1) & \Theta_{\theta_t}(\epsilon_2, \epsilon_2) & \cdots & \Theta_{\theta_t}(\epsilon_2, \epsilon_n) \\ \vdots & \vdots & \ddots & \vdots \\ \Theta_{\theta_t}(\epsilon_n, \epsilon_1) & \cdots & \cdots & \Theta_{\theta_t}(\epsilon_n, \epsilon_n) \end{bmatrix} \quad (110)$$

We then compute the Moore-Penrose pseudoinverse of this matrix with a certain degree of regularization²⁰ of the Gram matrix to numerically simulate the effect of inverting the continuous NTK integral operator.

Now that we have obtained the pseudoinverse NTK matrix, to numerically approximate the NTK-induced metric, we also need the velocity fields. We stack the velocity fields of n samples in an nd -dimensional column vector:

$$\mathbf{U}_t = \begin{pmatrix} f_{t+1}(\epsilon_1) - f_t(\epsilon_1) \\ f_{t+1}(\epsilon_2) - f_t(\epsilon_2) \\ \vdots \\ f_{t+1}(\epsilon_n) - f_t(\epsilon_n) \end{pmatrix} \quad (111)$$

The numerical approximation of the metric at each training step t will be:

$$\langle u, V \rangle_t = \langle u, \Theta^+ u \rangle_t \approx \mathbf{U}_t^\top \mathbf{K}_t^+ \mathbf{U}_t \quad (112)$$

Then, the experimental test comes in this: we wish to analyze, for each iteration, how close this approximation of the metric is to $\|\Delta\theta_t\|_2^2 = \|\theta_{t+1} - \theta_t\|_2^2$, the squared L2 norm of the parameter change that the underlying neural network actually underwent for that step. This comparison very strongly reveals how the actual change compares with the theoretically predicted change. To statistically analyze this, for each experiment, we use the following statistical measures between or on the two quantities over a training of 50 iterations: mean absolute relative error, median relative error, RMSE, mean NTK metric, and mean parameter L2 norm. For redundancy, we run this same procedure by varying various parameters. We decide to vary the following: learning rate for SGD optimizer, number of samples used to approximate the NTK metric (n mentioned

²⁰This is to solve the issue of PyTorch having small floating point eigenvalues, which will explode when the inverse is taken. More specifically, we first symmetrized the NTK matrix by doing $(K + K^\top)/2$ and then used the `torch.linalg.pinv()` function with `rtol` parameter set to $1e-10$.

above), number of hidden layers, and activation function used. For each of these, we vary the value for each quantity while holding others constant. When held constant, we set the above quantities to be, respectively, 10^{-2} , 20, 5, all sigmoid used throughout.

A sample output from an experiment would be in the likes of:

```
At step 20, NTK metric is approximately 0.0001936427050761048. The parameter perturbation L2 norm is 0.00019359122276440876
At step 21, NTK metric is approximately 0.00019627945145796432. The parameter perturbation L2 norm is 0.0001962213873754721
At step 22, NTK metric is approximately 0.00020014285328594954. The parameter perturbation L2 norm is 0.0002000770067906003
At step 23, NTK metric is approximately 0.0001918642742328854. The parameter perturbation L2 norm is 0.00019180125992749678
At step 24, NTK metric is approximately 0.00019152619412789987. The parameter perturbation L2 norm is 0.00019146030215373922
At step 25, NTK metric is approximately 0.00019838777725122886. The parameter perturbation L2 norm is 0.00019831368361915394
At step 26, NTK metric is approximately 0.0001918369320145866. The parameter perturbation L2 norm is 0.00019176439087752542
At step 27, NTK metric is approximately 0.00018903075842948759. The parameter perturbation L2 norm is 0.00018895614491937516
At step 28, NTK metric is approximately 0.00019402401886596998. The parameter perturbation L2 norm is 0.0001939434459560877
At step 29, NTK metric is approximately 0.00019714592034025748. The parameter perturbation L2 norm is 0.00019705848456589144
At step 30, NTK metric is approximately 0.00019189255220723142. The parameter perturbation L2 norm is 0.00019180482957123775
```

Now, with all the experiment details specified, let us present our results.²¹

Iteration 1: Learning Rate for Optimizer

Learning rate	Mean absolute relative error	Median relative error	RMSE	Mean NTK metric	Mean parameter norm ²
10^{-2}	0.03661%	0.03589%	8.3665×10^{-8}	1.9184833×10^{-4}	1.9178105×10^{-4}
10^{-3}	0.03157%	0.02798%	7.8503×10^{-8}	2.2066349×10^{-4}	2.2061008×10^{-4}
10^{-4}	0.004738%	0.004754%	1.1242×10^{-10}	2.3656510×10^{-6}	2.3657631×10^{-6}
5×10^{-5}	0.0001216%	0.0001168%	8.5219×10^{-13}	5.9717995×10^{-7}	5.9717927×10^{-7}

Table 1: Iteration 1

Our first test of changing the learning rate of the SGD optimizer while holding all other parameters constant cannot reveal the relationship between the two quantities clearer. First, we see that, over all 50 training iterations, the NTK-metric approximated by the Gram matrix and stacked velocity fields and the actual L2 norm of parameter change are extremely close, equal up to 2~3 decimal places, with no outliers. It is also very clear that the smaller the learning rate is, the more the mean absolute and relative error between the two quantities strongly decrease. This can be easily explained by the fact that a smaller learning rate would mean a smaller step taken in the direction of the gradient, which would make the finite approximation $u_t(f_\theta(\epsilon_i)) \approx f_{t+1}(\epsilon_i) - f_t(\epsilon_i)$ more granular and thus more accurate. Also notice that the means of both quantities decrease steadily as the learning rate is made smaller. This is also natural, since a smaller learning rate means a smaller step size, yielding smaller changes in the parameter and smaller velocity fields. This is already very strong empirical evidence that the NTK-metric does indeed correctly represent the parameter level change in the underlying neural architecture.

Iteration 2: n , the number of samples

This experiment also is very clear: unsurprisingly, increasing n tends to decrease the median absolute and relative error. The insight here also is clear: the more samples there are, the more representative both quantities are. Since the prior distribution being drawn from is the Gaussian distribution, with a very small number of samples, the samples may differ substantially, causing the parameter perturbation and the NTK metric to be evaluated on different sample configurations. The more samples there are, the more representative both quantities become. But also note that, even in the extremely coarse case where $n = 1$, the mean error between the two quantities is still very small at an absolute scale, being, respectively, around 0.5% and 0.4%.

²¹See <https://github.com/ufofVincent/Manifold-of-Realizable-Distributions-Research> for code and full results for all 50 steps for each of the following runs, stored as .txt files.

n	Mean absolute relative error	Median relative error	RMSE	Mean NTK metric	Mean parameter norm ²
1	0.5069%	0.4056%	7.4487×10^{-5}	6.85614×10^{-3}	6.82236×10^{-3}
5	0.4322%	0.3409%	7.7629×10^{-5}	5.45053×10^{-3}	5.39800×10^{-3}
10	0.4006%	0.3185%	7.4050×10^{-5}	5.42348×10^{-3}	5.37795×10^{-3}
20	0.3736%	0.3104%	5.3858×10^{-5}	6.29991×10^{-3}	6.27368×10^{-3}
30	0.4492%	0.3363%	7.1927×10^{-5}	6.40308×10^{-3}	6.36260×10^{-3}

Table 2: Iteration 2

This also is very strong evidence of the conceptual correlation between the two quantities, since even a single sample can reveal a strong correlation between the two quantities.

Iteration 3: The number of hidden layers

n	Mean absolute relative error	Median relative error	RMSE	Mean NTK metric	Mean parameter norm ²
1	0.3210%	0.2549%	6.1435×10^{-5}	5.86841×10^{-3}	5.85313×10^{-3}
3	0.5889%	0.4893%	7.9031×10^{-5}	5.60238×10^{-3}	5.55125×10^{-3}
5	0.4105%	0.2936%	8.4316×10^{-5}	4.98210×10^{-3}	4.92778×10^{-3}
10	0.4316%	0.3218%	7.3935×10^{-5}	5.73848×10^{-3}	5.69228×10^{-3}
20	0.3812%	0.3035%	4.8689×10^{-5}	5.06113×10^{-3}	5.03295×10^{-3}

Table 3: Iteration 3

For this iteration, we must be much more conservative and careful in our assessment of the relationship between the number of hidden layers and the predictive power of the NTK metric, since in this iteration, we notice there is no clear or straightforward correlation of the kind observed in the previous two experiments. The mean absolute relative error first increases but then steadily decreases as the neural network becomes deeper. Overall, we observe a relatively stable agreement between the two quantities. We notice that deeper neural networks do not necessarily jeopardize the predictive power of NTK’s, meaning that the correspondence between the two is robust and is not confined to merely shallow networks.

Iteration 4: Activation function

Activation function	Mean absolute relative error	Median relative error	RMSE	Mean NTK metric	Mean parameter norm ²
All Sigmoid	0.5775%	0.4740%	8.2058×10^{-5}	6.09942×10^{-3}	6.04750×10^{-3}
Alternating ReLU/Sigmoid	17.71%	0.3885%	1.384×10^{-2}	9.9440×10^{-3}	6.0687×10^{-3}
All ReLU	672.2%	171.4%	6.789×10^{-1}	2.083×10^{-1}	5.133×10^{-3}

Table 4: Iteration 4

The activation-function experiment produces the largest discrepancy observed in our tests. The all-sigmoid network retains close agreement between the empirical NTK quadratic form and the squared parameter displacement, whereas the ReLU-containing networks exhibit occasional—and, in the all-ReLU case, extensive—departures from this agreement. In particular, the difference between the median and mean relative errors for the alternating ReLU/Sigmoid network indicates that a comparatively small number of large spikes substantially affect the mean.

These results are consistent with the finite-step linearization becoming less accurate when an update crosses activation boundaries. However, they do not by themselves identify activation-boundary crossing as the unique cause of the discrepancy. Numerical conditioning of the sampled NTK matrix, changes in its effective rank under the pseudoinverse cutoff, and amplification of finite-difference errors may also contribute. The ReLU

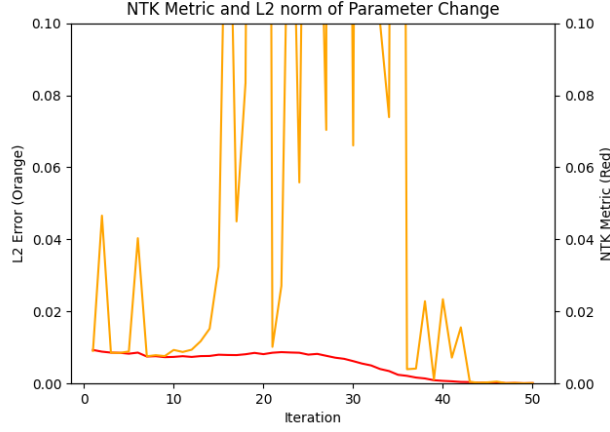


Figure 4: Plot of Approximated NTK-Metric (Orange) and L2 norm of parameter change (Red) against 50 Training Iterations for the all ReLU run

experiment should therefore be interpreted as evidence of instability in the present finite-step approximation, rather than as conclusive evidence that the instantaneous NTK identity fails for every nonsmooth architecture.

To make this distinction precise, let J_t denote the sampled Jacobian at θ_t , let

$$K_t = J_t J_t^\top, \quad (113)$$

and define

$$U_t = f_{\theta_t + \Delta\theta_t} - f_{\theta_t} = J_t \Delta\theta_t + R_t, \quad (114)$$

where R_t is the finite-step linearization remainder evaluated on the sampled latent points. The empirical NTK quadratic form is then

$$M_t = U_t^\top K_t^+ U_t. \quad (115)$$

Substituting the decomposition of U_t gives

$$M_t = \Delta\theta_t^\top J_t^\top K_t^+ J_t \Delta\theta_t + 2R_t^\top K_t^+ J_t \Delta\theta_t + R_t^\top K_t^+ R_t. \quad (116)$$

Because

$$J_t^\top (J_t J_t^\top)^+ J_t = P_{\text{Range}(J_t^\top)}, \quad (117)$$

the first term satisfies

$$\Delta\theta_t^\top J_t^\top K_t^+ J_t \Delta\theta_t = \left\| P_{\text{Range}(J_t^\top)} \Delta\theta_t \right\|_2^2 \leq \|\Delta\theta_t\|_2^2. \quad (118)$$

Consequently,

$$\begin{aligned} M_t - \|\Delta\theta_t\|_2^2 = & - \|P_{\ker J_t} \Delta\theta_t\|_2^2 \\ & + 2R_t^\top K_t^+ J_t \Delta\theta_t + \underbrace{R_t^\top K_t^+ R_t}_{\geq 0 \text{ due to positive (semi)-definiteness}}. \end{aligned} \quad (119)$$

This decomposition explains why both overestimation and underestimation are possible. The projection term is nonpositive, the quadratic remainder term is nonnegative, and the cross term has no fixed sign. Therefore, the predominance of overestimation, as shown in Tables 5 and 6, where we display, for all runs above, the frequency of overestimation and underestimation, is an empirical property of the present runs; it does not follow automatically from the existence of a Taylor remainder.

Outcome	Count	Percentage
$M_{\text{NTK}} > \ \Delta\theta\ _2^2$	754	88.71%
$M_{\text{NTK}} < \ \Delta\theta\ _2^2$	96	11.29%

Table 5: Frequency of overestimation and underestimation of the squared parameter perturbation by the empirical NTK metric across 850 training steps.

Variation	Overestimation	Underestimation	Total
Learning rate	126	74	200
Sample count n	241	9	250
Hidden layers	247	3	250
Activation Function	140	10	150
Total	754	96	850

Table 6: Breakdown of NTK-metric overestimation and underestimation by experimental variation.

The pseudoinverse also makes the comparison sensitive to conditioning. If J_t has a small nonzero singular value, then K_t^+ places a large weight on the corresponding output-space direction. A relatively small component of R_t in such a direction can consequently produce a large value of $R_t^\top K_t^+ R_t$. The spikes in the ReLU experiments may therefore reflect an interaction among activation-pattern changes, finite-step linearization error, and the spectrum and numerical rank of the sampled Jacobian.

There is also an important reason why close agreement is expected in the smooth experiments. When plain SGD is computed from the same sampled outputs used to construct J_t , its update generally has the form

$$\Delta\theta_t = J_t^\top a_t \quad (120)$$

for some sampled output-space vector a_t . Hence

$$\Delta\theta_t \in \text{Range}(J_t^\top), \quad (121)$$

and the exactly linearized quadratic form satisfies

$$(J_t \Delta\theta_t)^\top K_t^+ (J_t \Delta\theta_t) = \|\Delta\theta_t\|_2^2. \quad (122)$$

Thus, the experiment is best understood as testing how accurately the finite output displacement U_t is represented by $J_t \Delta\theta_t$, together with the numerical stability of the sampled pseudoinverse. It is not an independent test of the density-manifold construction or of the pushforward differential in the ambient distribution space.

Within this interpretation, the results are encouraging. The smooth-network runs exhibit small finite-step discrepancies across the tested learning rates, sample counts, and depths, while smaller learning rates generally improve the approximation. The ReLU runs reveal that the same computation can become unstable when finite-step nonlinearities and pseudoinverse sensitivity are more pronounced. These experiments therefore support the numerical consistency of the NTK-pseudoinverse representation in the tested smooth settings, while also identifying finite-step linearization and spectral conditioning as important limitations requiring further investigation. We welcome more types of experiments not just on the validity of numerical approximation on the NTK-metric but also on the manifold construction itself as future work.

7 Conclusion

In this paper, we successfully constructed a geometric framework for neural architecture guided distribution evolution using the language of Riemannian manifolds. We first noticed the mediating role of the neural tangent kernel in drift field guided training, which paves the foundation to the insight that it acts as a mobility tensor. We then induced the set of local manifolds of distributions reachable by a neural architecture by constructing charts to a Hilbert manifold acting as distribution coordinates and using the Constant Rank Theorem to induce that topological space. We then showed, through the conceptual tools of Onsager geometry, how the pseudoinverse of the neural tangent kernel represents a Riemannian metric on the graph-induced submanifold on the product manifold between parameter space and the distribution manifold. We discussed that the graph construction is not merely a device for defining a metric: it supplies the global manifold structure that the set-theoretic realization locus may lack because of self-intersections, singularities, and nonidentifiable parameterizations. Conceptually, what we do here is elegant: we have shown that there is a space capable of being characterized by differential geometry and topology that neural realizable architecture-induced probability distributions obey, and that existing work in relevant fields of mathematics, such as differential geometry and topology, can be seamlessly applied to analyze such a space. Differential geometry and neural architecture guided distribution evolution are, thus, integrated into a coherent conceptual picture, opening future work on how this relation can be further refined and applied.

A natural benefit that would follow is that many existing concepts in mathematics, geometry, and physics would easily map over to existing works in machine learning. For example, the notion of geodesics, which, in physics, refers to the generalized notion of a straight line on a curved manifold, easily can be borrowed to refer to the shortest and thus least expensive way, at least at an architectural parameter level, to realize a particular probability transport. Our construction here also rigorously quantifies and characterizes the notion of neural realizability: instead of beginning from a manifold and consider what distribution evolutions it can realize, one can likewise consider the question of, given a distribution evolution, whether any architecture could realize it. By showing that the former has a geometric basis, the latter is a concrete application thereof that could reframe how distribution evolution is realized.

Additionally, one specific application of this framework we present is the geometric view of expressibility of different neural architectures. By demonstrating how the manifold immersion can be constructed through a sample pair of neural architectures, which in this case is a convolutional neural network and a Transformer, these results suggest a branch-aware geometric approach to neural-architecture expressivity: architectures may be compared through maps between their parameter–distribution graphs and through inclusions among the distributional deformations generated by their realization differentials.

8 Limitations and Future Work

8.1 Topology of the Regular Parameter Space

One construction that our paper has relied on, to take full advantage of the Constant Rank theorem, is the regular parameter space, as mentioned in section 4.1.5: the set of parameters under which the differential of the pushforward operator $\Phi : \mathbb{R}^p \rightarrow P^\infty(\mathbb{R}^d)$ is injective. In our discussion in that section, we mentioned that in most cases, most parameters should meet that criterion, since the probability of the differential not being full rank is small but not negligible. In particular, especially for an overparametrized neural architecture, where $p \gg d$, the operator

$$u = J_\theta f \dot{\theta} \tag{123}$$

is certainly non-injective, insofar as there must exist nonzero $\dot{\theta}$ such that $J_\theta f \dot{\theta} = 0$, or equivalently, $J_\theta f$ having a nontrivial kernel. The velocity field produced by this operator also may be q-divergence free, meaning $-\nabla \cdot (qu) = 0$. So, we admit that the regular parameter set certainly is not equal to the parameter space as a whole, since there are two layers that may result in a distribution deformation of 0, and thus equivalently noninjectivity.

We wish to quantify this notion. A natural question that would emerge then is, what is the topology of this regular parameter set $\mathbb{R}_{\text{reg}}^p$ in relation to the entire parameter space $\mathbb{R}^p$? From a topological perspective, we have already proven that it is an open subset, i.e., every point has a neighborhood sufficiently small that is contained within the set, but we additionally conjecture that the set of regular parameters is a **dense** subset of the total Euclidean space of all parameters. And, from a measure-theoretic perspective, is the set that is not in the regular parameter set a null set, i.e., $\text{Leb}(\mathbb{R}^p \setminus \mathbb{R}_{\text{reg}}^p) = 0$? It is highly desirable to prove this conjecture, since if the regular parameter set is dense, one could introduce, by the definition of a dense subset, a notion of "regularization": for a parameter not in the regular parameter set, one can always find a parameter in the regular parameter set that is arbitrarily close to it under a given norm: $\forall \theta \notin \mathbb{R}_{\text{reg}}, \forall \epsilon > 0, \exists \theta_{\text{reg}} \in \mathbb{R}_{\text{reg}}^p \text{ s.t. } \|\theta - \theta_{\text{reg}}\| \leq \epsilon$. That way, even if a parameter is not regular, we can substitute it via an arbitrarily close parameter that would induce an essentially equal distribution.

This question can also be of not merely topological or mathematical interest, but also could reveal insights on the concrete properties of an architecture, since an overparametrized neural architecture necessarily would have parameter redundancies. Another question one then can think of is, how can one change the configurations of the neural network so that this subset becomes dense. So a question we pose would be *under what architectural conditions would the regular parameter space be dense?* This could open a new avenue of exciting research.

8.2 Alternative Solutions to the Parameter Identifiability / Global Noninjectivity of the Pushforward Φ Problem

We have discussed thoroughly in section 4.2 the reason why we used the graph product submanifold construction: to both repair the lack of a global manifold structure on the distribution locus and to circumvent the issue of multiple parameters giving rise to the same distribution via a neural architecture given pushforward, i.e., the lack of global injectivity of the pushforward Φ . It would be helpful to discuss other solutions the author of this paper has considered, since it may lead to alternative formulations of what this paper conceptually advances. The initial solution was **quotient geometry**: instead of explicitly grouping every distribution with the parameter that give rise to it, we can simply, in the regular distribution space, quotient out parameters that give rise to the same distribution. In that case, the underlying parameter space will be $\mathbb{R}_{\text{reg}}^p / \sim$. This method would easily circumvent the problem, by simply grouping parameters together and collapsing them under one object: $[\theta] \in \mathbb{R}_{\text{reg}}^p / \sim$, where $[\theta] = \{\eta \in \mathbb{R}_{\text{reg}}^p \mid \Phi(\eta) = \Phi(\theta)\}$. This would be a solution that abstracts the problem away by abandoning the variety of geometries different parameters would locally have and selecting only one to represent them all. In fact, in existing ML literature, the study of that quotient space is explored, especially in relation to concepts such as parameter symmetries, topology, and neuromanifolds Zhao et al. [2025], Alvarado [2026]. This was a solution considered by the author, but we devised and preferred the graph solution since the latter preserves the variety of parameters and thus their local geometries. We also further believe that quotient geometry may have other obstructions that made the graph solution stand out.

Another interesting solution, one that is also supported by existing concepts from differential geometry, is **fiber bundles / fibered spaces**. We can let $\mathcal{M}_f^{\text{prior}}$, a topological space, be the base space in a fiber bundle, and the fiber at each point would be the set of all parameters that give rise to the distribution that point represents: $F|_{q_\theta} = \Phi^{-1}(q_\theta) = \{\theta \in \mathbb{R}_{\text{reg}}^p \mid \Phi(\theta) = q\}$. Locally then, we will have $\pi^{-1}(U) \cong U \times F$. This construction is certainly not without its merits, and may in fact, admittedly, capture the parameter-distribution relation in a conceptually more intuitive way than what we presented in this paper, since the number of parameters that give rise to one distribution certainly varies. We encourage readers of this paper to experiment on these two solutions and propose new ones, which would lead to interest comparisons as to the merits of each solution.

It also should be remarked that another way to circumvent the problem is to further restrict the space of parameters to not just the differential of the pushforward being injective, but also the pushforward being injective, yielding a hierarchy in the likes of $\mathbb{R}_{\text{inj}}^p \subset \mathbb{R}_{\text{reg}}^p \subset \mathbb{R}^p$. This would be a more austere solution, but it can immediately be guaranteed that $\Phi(\mathbb{R}_{\text{inj}}^p)$ is an immersed manifold as a whole. This solution would entirely not require the graph construction, but we believe it would greatly reduce the size of the manifold to the extent of damaging the expressivity of the neural architecture it represents, given that parameter redundancy is unavoidable for overparametrized architectures.

8.3 Alternative Metrics on Parameter-Distribution Graph

The current choice of the metric presented in this paper ultimately measures the Euclidean metric on the parameter perturbations required to realize the corresponding distributional deformations, requiring the NTK to express on the distributional level, instead of merely the intrinsic geometric distance of those two deformations on the distribution space. In other words, we are not measuring simply how different the two deformations are geometrically, but the neural network required effort. The choice of metric, thus, would give rise to different conceptual interpretations on what the real number outputted by the metric would represent. This paper invites the readers to experiment with other metrics, which would produce different notions of distance, angle, and length.

Unsurprisingly, in relevant areas of literature, there are already many existing metrics that could be borrowed over. As mentioned in the related works section, the **Wasserstein metric** would be a canonical and well known metric that could replace our metric:

$$g_q(\dot{\mathbf{q}}, \dot{\mathbf{q}}) = \inf_{u \in \mathcal{V} \text{ s.t. } \dot{q} = -\nabla(q_\theta u)} \int \|u(x)\|^2 q_\theta(x) dx \quad (124)$$

Another candidate would be the **Fisher-Rao Metric**, a metric from information theory that is likewise borrowed into deep learning Liang et al. [2019]:

$$g_q(\dot{\mathbf{q}}_1, \dot{\mathbf{q}}_2) = \int \frac{\dot{q}_1(x)\dot{q}_2(x)}{q(x)} dx \quad (125)$$

Alternatively, since the regular parameter space we constructed makes use of Sobolev space for its coordinates, we could also borrow the **negative Sobolev metric**, which is a metric that already exists in mathematical literature:

$$g_q(\dot{\mathbf{q}}_1, \dot{\mathbf{q}}_2) = \int_{\mathbb{R}^d} (1 + \|\xi\|^2)^{-s} \widehat{\dot{q}_1(\xi)} \overline{\widehat{\dot{q}_2(\xi)}} d\xi \quad (126)$$

This would be a metric that measures distributional deformations after smoothing high-frequency components, a description of flat ambient Hilbert geometry rather than transport geometry.

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9 Appendix

9.1 Assumptions Section

Definition 24 (Uniformly square-integrable parameter Jacobian).

$$\sup_{\theta \in K} \mathbb{E}_{\epsilon \sim p_{\text{prior}}} \|J_{\theta} F_{\theta}(\epsilon)\|_F^2 < \infty, \quad (127)$$

where K is a compact space in the parameter space $\mathbb{R}^p$ and $\|\cdot\|_F$ denotes the Frobenius norm for matrices.

Remark This is a local condition of regularity that can be assumed for most neural architectures, and is a crucial assumption that our paper relies on, especially in the later discussion on the boundedness of the NTK as a mobility tensor. A natural question that would emerge is how often this assumptions actually is satisfied by neural architectures. We will answer that most neural architectures, such as MLPs with ReLU or smooth activation functions, CNN's, Transformers, all satisfy it, when provided that the distribution p_{prior} sampled from has finite moments. Generally, a finite second moment would suffice. And especially since the distribution we choose is the Gaussian distribution, which has finite moments of all orders: $\mathbb{E}_{\epsilon \sim \mathcal{N}(0, \mathbb{I})} [\|\epsilon\|^n] < \infty \forall n \in \mathbb{N}$.

The reason why this assumption is needed is to ensure that the Jacobian of the neural architecture acts as a valid operator into L^2 and for its adjoint to exist and be bounded. Another useful corollary of this assumption is this additional guarantee:

Corollary 25 (The NTK as an integral operator is Hilbert-Schmidt).

$$\int_X \int_X \|\Theta_{\theta}(\epsilon, \epsilon')\|_F^2 dp_{\text{prior}}(\epsilon) dp_{\text{prior}}(\epsilon') < \infty \quad (128)$$

Proof. This result directly follows from the uniformly square-integrable parameter Jacobian assumption we made in the beginning of this paper. The proof is relatively straightforward: simply observe a crucial property of the Frobenius norm:

$$\|AB^T\|_F \leq \|A\|_F \|B\|_F$$

So,

$$\|\Theta_{\theta}(\epsilon, \epsilon')\|_F^2 \leq \|J_{\theta}(\epsilon)\|_F^2 \|J_{\theta}(\epsilon')\|_F^2$$

Integrating over X weighed by the distribution μ , or, equivalently, taking the expectation on both sides,

$$\mathbb{E}_{(\epsilon, \epsilon') \sim \mu \oplus \mu'} [\|\Theta_{\theta}(\epsilon, \epsilon')\|_F^2] \leq \mathbb{E}_{\epsilon, \epsilon'} [\|J_{\theta}(\epsilon)\|^2 \|J_{\theta}(\epsilon')\|^2]$$

Since the two samplings are independent, we have

$$\mathbb{E}_{(\epsilon, \epsilon') \sim \mu \oplus \mu'} [\|\Theta_{\theta}(\epsilon, \epsilon')\|_F^2] \leq (\mathbb{E}_{\epsilon} [\|J_{\theta}(\epsilon)\|^2])^2$$

Now we may use the square-integrable parameter Jacobian assumption. From that, we know the term on the right is bounded. So,

$$\mathbb{E}_{(\epsilon, \epsilon') \sim \mu \oplus \mu'} [\|\Theta_{\theta}(\epsilon, \epsilon')\|_F^2] \leq (\mathbb{E}_{\epsilon} [\|J_{\theta}(\epsilon)\|^2])^2 < \infty$$

□

9.2 Existence of the Potential Representation for Distributional Deformations

Let $q \in C^\infty(M)$ be a strictly positive probability density on a smooth, connected, closed Riemannian manifold M , and let $u \in \mathfrak{X}(M)$ be a smooth velocity field. Consider the distributional deformation induced by u through the continuity equation,

$$h := -\nabla \cdot (qu). \quad (129)$$

We seek a scalar potential $\phi \in C^\infty(M)$ such that the gradient velocity field $\nabla\phi$ induces the same distributional deformation:

$$-\nabla \cdot (q\nabla\phi) = -\nabla \cdot (qu). \quad (130)$$

Importantly, Eq. (130) does not require $\nabla\phi = u$ pointwise. It requires only that the two velocity fields induce the same infinitesimal deformation of the probability density.

Define the weighted elliptic operator

$$L_q\phi := -\nabla \cdot (q\nabla\phi). \quad (131)$$

Equation (130) is therefore the weighted Poisson equation

$$L_q\phi = h. \quad (132)$$

A necessary and sufficient compatibility condition for the existence of a solution of a Poisson equation on a closed manifold is

$$\int_M h \, d\text{vol}_M = 0. \quad (133)$$

For a deformation generated by the continuity equation, this condition holds automatically. Indeed, by the divergence theorem and the assumption that M has no boundary, i.e. $\partial M = \emptyset$,

$$\int_M h \, d\text{vol}_M = - \int_M \nabla \cdot (qu) \, d\text{vol}_M \quad (134)$$

$$= 0. \quad (135)$$

Thus every smooth deformation arising from a smooth velocity field through the continuity equation satisfies the compatibility condition required by the weighted Poisson equation. Geometrically, Eq. (133) expresses conservation of total probability:

$$\frac{d}{dt} \int_M q_t \, d\text{vol}_M = \int_M \dot{q}_t \, d\text{vol}_M = 0. \quad (136)$$

This can also be understood as follows: the distributional deformation must not change the total probability across all points on the space $\mathbb{R}^d$, and so its integral over the entire space must equal 0. This preserves the total mass of probability at 1. This is a result we have arrived at and stressed multiple times in the main content of this paper.

The solution ϕ is unique only modulo the addition of a spatially constant function. If ϕ solves Eq. (130), then for any $C \in \mathbb{R}$,

$$\nabla(\phi + C) = \nabla\phi, \quad (137)$$

and consequently

$$L_q(\phi + C) = L_q\phi. \quad (138)$$

This motivates the identification of potentials modulo constants,

$$\phi \in C^\infty(M)/\mathbb{R}. \quad (139)$$

So, all this means that any regular distributional deformation (we define "regular" to be one that satisfies equation 133) admits at least one velocity field that realizes it.

Equivalently, uniqueness may be imposed by choosing a normalization such as

$$\int_M \phi q \, d\text{vol}_M = 0. \quad (140)$$

The strict positivity of q plays an essential role in this construction. Since $q \in C^\infty(M)$, $q > 0$, and M is compact, there exists a constant $c > 0$ such that

$$q(x) \geq c \quad \text{for all } x \in M. \quad (141)$$

Consequently, L_q is a uniformly elliptic second-order differential operator. Moreover, on a connected closed manifold its kernel consists precisely of the constant functions:

$$\ker L_q = \mathbb{R}. \quad (142)$$

Standard elliptic theory then implies that the range of L_q consists of the smooth zero-mean functions,

$$\text{Range}(L_q) = \left\{ h \in C^\infty(M) : \int_M h \, d\text{vol}_M = 0 \right\}. \quad (143)$$

Hence, for every smooth mass-preserving distributional deformation h , there exists a smooth potential ϕ , unique modulo constants, satisfying

$$-\nabla \cdot (q \nabla \phi) = h. \quad (144)$$

In particular, for the NTK-mediated velocity field

$$u_\theta = \Theta_\theta \mathbf{V}_{p,q}, \quad (145)$$

the induced distributional deformation

$$\dot{q}_\theta = -\nabla \cdot (q_\theta \Theta_\theta \mathbf{V}_{p,q}) \quad (146)$$

admits a potential representation

$$\dot{q}_\theta = -\nabla \cdot (q_\theta \nabla \phi_\theta) \quad (147)$$

for some $\phi_\theta \in C^\infty(M)/\mathbb{R}$, provided the preceding smoothness assumptions hold. Equivalently,

$$\nabla \cdot [q_\theta (\Theta_\theta \mathbf{V}_{p,q} - \nabla \phi_\theta)] = 0. \quad (148)$$

Thus $\Theta_\theta \mathbf{V}_{p,q}$ need not itself be a gradient field; rather, it is equivalent at the level of infinitesimal distributional deformation to the gradient representative $\nabla \phi_\theta$. In fact, there is theoretical results that demonstrate that drift fields themselves in general are not conservative Franz et al. [2026]. Weighing it by the NTK may or may not change its conservativity, but the above results stand.

9.3 Drift Fields are L^2 integrable

A helpful side lemma that should be proven, since we have asserted that the NTK is an operator from $\Theta_\theta : L^2(p_{\text{prior}}; \mathbb{R}^d) \rightarrow L^2(p_{\text{prior}}; \mathbb{R}^d)$, is to show that drift fields as defined by Deng et al. [2026] are L^2 integrable. Equivalently, by letting $x = f_\theta(\epsilon)$, weighing ϵ by p_{prior} is equivalent to weighing x by $q(x)$. That is,

$$\int_{\mathbb{R}^d} \|\mathbf{V}_{p,q}(x)\|^2 q(x) dx < \infty \quad (149)$$

We first will prove that the choice of the Gaussian kernel function as used by Deng et al. and as we also have used in this paper will lead to an easy proof. But also have discovered a general sufficient condition for any drift field to be L^2 integrable.

Proposition 26 (The drift field using the Gaussian kernel with any $\tau > 0$ is L^2 integrable).

Proof. Let the kernel be the Gaussian kernel

$$k(x, y) = e^{-\frac{1}{\tau} \|x - y\|_2^2}.$$

Starting from

$$\int_{\mathbb{R}^d} \left\| \frac{1}{Z_p Z_q} \mathbb{E}_{y^+ \sim p, y^- \sim q} [k(x, y^-) k(x, y^+) (y^+ - y^-)] \right\|_2^2 dx,$$

expanding the expectation gives

$$\int_{\mathbb{R}^d} \left\| \frac{1}{Z_p Z_q} \int_{\mathbb{R}^d} \int_{\mathbb{R}^d} e^{-\frac{1}{\tau} \|x-y^-\|_2^2} e^{-\frac{1}{\tau} \|x-y^+\|_2^2} (y^+ - y^-) p(y^+) q(y^-) dy^+ dy^- \right\|_2^2 dx.$$

By Jensen's inequality,

$$\left\| \mathbb{E}_{y^+ \sim p, y^- \sim q} [k(x, y^-) k(x, y^+) (y^+ - y^-)] \right\|_2^2 \leq \mathbb{E}_{y^+ \sim p, y^- \sim q} [k(x, y^-)^2 k(x, y^+)^2 \|y^+ - y^-\|_2^2].$$

Hence,

$$\begin{aligned} & \int_{\mathbb{R}^d} \left\| \frac{1}{Z_p Z_q} \mathbb{E}_{y^+ \sim p, y^- \sim q} [k(x, y^-) k(x, y^+) (y^+ - y^-)] \right\|_2^2 dx \\ & \leq \frac{1}{Z_p^2 Z_q^2} \int_{\mathbb{R}^d} \int_{\mathbb{R}^d} \int_{\mathbb{R}^d} e^{-\frac{2}{\tau} \|x-y^-\|_2^2} e^{-\frac{2}{\tau} \|x-y^+\|_2^2} \|y^+ - y^-\|_2^2 p(y^+) q(y^-) dy^+ dy^- dx. \end{aligned}$$

By Tonelli's theorem, we may change the order of integration:

$$\begin{aligned} & = \frac{1}{Z_p^2 Z_q^2} \int_{\mathbb{R}^d} \int_{\mathbb{R}^d} \|y^+ - y^-\|_2^2 p(y^+) q(y^-) \\ & \quad \cdot \left(\int_{\mathbb{R}^d} e^{-\frac{2}{\tau} \|x-y^-\|_2^2} e^{-\frac{2}{\tau} \|x-y^+\|_2^2} dx \right) dy^+ dy^-. \end{aligned}$$

Set

$$m = \frac{y^+ + y^-}{2}.$$

Using the identity

$$\|x - y^+\|_2^2 + \|x - y^-\|_2^2 = 2 \|x - m\|_2^2 + \frac{1}{2} \|y^+ - y^-\|_2^2,$$

we obtain

$$\begin{aligned} & \frac{2}{\tau} \|x - y^+\|_2^2 + \frac{2}{\tau} \|x - y^-\|_2^2 \\ & = \frac{4}{\tau} \|x - m\|_2^2 + \frac{1}{\tau} \|y^+ - y^-\|_2^2. \end{aligned}$$

Therefore,

$$\begin{aligned} & \int_{\mathbb{R}^d} e^{-\frac{2}{\tau} \|x-y^-\|_2^2} e^{-\frac{2}{\tau} \|x-y^+\|_2^2} dx \\ & = e^{-\frac{1}{\tau} \|y^+ - y^-\|_2^2} \int_{\mathbb{R}^d} e^{-\frac{4}{\tau} \|x-m\|_2^2} dx \\ & = \left(\frac{\pi\tau}{4} \right)^{d/2} e^{-\frac{1}{\tau} \|y^+ - y^-\|_2^2}. \end{aligned}$$

It follows that

$$\|\mathbf{V}_{p,q}\|_{L^2}^2 \leq \frac{1}{Z_p^2 Z_q^2} \left(\frac{\pi\tau}{4} \right)^{d/2} \int_{\mathbb{R}^d} \int_{\mathbb{R}^d} \|y^+ - y^-\|_2^2 e^{-\frac{1}{\tau} \|y^+ - y^-\|_2^2} p(y^+) q(y^-) dy^+ dy^-.$$

Now the scalar function

$$r \longmapsto r^2 e^{-r^2/\tau}$$

is bounded on $[0, \infty)$. In particular,

$$r^2 e^{-r^2/\tau} \leq \frac{\tau}{e}.$$

Thus

$$\begin{aligned}\|\mathbf{V}_{p,q}\|_{L^2}^2 &\leq \frac{1}{Z_p^2 Z_q^2} \left(\frac{\pi}{4}\right)^{d/2} \frac{\tau}{e} \int_{\mathbb{R}^d} p(y^+) dy^+ \int_{\mathbb{R}^d} q(y^-) dy^- \\ &= \frac{\tau}{e Z_p^2 Z_q^2} \left(\frac{\pi}{4}\right)^{d/2} < \infty,\end{aligned}$$

since p and q are probability densities, whose integral would evaluate to 1. Hence

$$\boxed{\mathbf{V}_{p,q} \in L^2(\mathbb{R}^d; \mathbb{R}^d)}.$$

□

We also have a general sufficient condition under which any drift field is L^2 integrable, not just ones that use Gaussian kernels. We notice that p, q having finite second moment and the normalized kernels being bounded will guarantee membership in $L^2(q; \mathbb{R}^d)$. These are relatively mild assumptions, since, as discussed in section 2, the probability distributions this paper are concerned with are a subset of probability distributions with finite second moment. Bounded kernel functions are likewise in abundance.

Proposition 27. *Let $p, q \in \mathcal{P}_2(\mathbb{R}^d)$, and define the normalized kernels*

$$\tilde{k}_p(x, y) = \frac{k(x, y)}{Z_p(x)}, \quad \tilde{k}_q(x, y) = \frac{k(x, y)}{Z_q(x)}. \quad (150)$$

Suppose that there exists a constant $C > 0$ such that

$$|\tilde{k}_p(x, y)| \leq C, \quad |\tilde{k}_q(x, y)| \leq C \quad (151)$$

for all $x, y \in \mathbb{R}^d$. Then the drifting field

$$\mathbf{V}_{p,q}(x) = \mathbb{E}_{y^+ \sim p, y^- \sim q} \left[\tilde{k}_p(x, y^+) \tilde{k}_q(x, y^-) (y^+ - y^-) \right] \quad (152)$$

belongs to $L^2(q; \mathbb{R}^d)$.

Proof. By definition,

$$\|\mathbf{V}_{p,q}\|_{L^2(q)}^2 = \int_{\mathbb{R}^d} \|\mathbf{V}_{p,q}(x)\|_2^2 q(x) dx.$$

For each $x \in \mathbb{R}^d$, Jensen's inequality gives

$$\begin{aligned}\|\mathbf{V}_{p,q}(x)\|_2^2 &= \left\| \mathbb{E}_{y^+ \sim p, y^- \sim q} \left[\tilde{k}_p(x, y^+) \tilde{k}_q(x, y^-) (y^+ - y^-) \right] \right\|_2^2 \\ &\leq \mathbb{E}_{y^+ \sim p, y^- \sim q} \left[\left\| \tilde{k}_p(x, y^+) \tilde{k}_q(x, y^-) (y^+ - y^-) \right\|_2^2 \right].\end{aligned}$$

Using the boundedness of the normalized kernels,

$$\|\mathbf{V}_{p,q}(x)\|_2^2 \leq C^4 \mathbb{E}_{y^+ \sim p, y^- \sim q} [\|y^+ - y^-\|_2^2].$$

By the elementary inequality

$$\|y^+ - y^-\|_2^2 \leq 2\|y^+\|_2^2 + 2\|y^-\|_2^2,$$

we obtain

$$\|\mathbf{V}_{p,q}(x)\|_2^2 \leq 2C^4 (\mathbb{E}_{y^+ \sim p} \|y^+\|_2^2 + \mathbb{E}_{y^- \sim q} \|y^-\|_2^2).$$

Since $p, q \in \mathcal{P}_2(\mathbb{R}^d)$, both second moments are finite. Hence there exists a constant $M < \infty$, independent of x , such that

$$\|\mathbf{V}_{p,q}(x)\|_2^2 \leq M.$$

Therefore,

$$\begin{aligned}\|\mathbf{V}_{p,q}\|_{L^2(q)}^2 &= \int_{\mathbb{R}^d} \|\mathbf{V}_{p,q}(x)\|_2^2 q(x) dx \\ &\leq M \int_{\mathbb{R}^d} q(x) dx \\ &= M < \infty,\end{aligned}$$

because q is a probability density, and thus its integral over the entire output space is 1. Thus

$$\mathbf{V}_{p,q}(x) \in L^2(q; \mathbb{R}^d).$$

Or equivalently,

$$\mathbf{V}_{p,q}(f_\theta(\epsilon)) \in L^2(p_{\text{prior}}; \mathbb{R}^d).$$

□

9.4 Nonatomic, smooth, positive distributions generated by neural architectures

The smooth-manifold formulation of the space of probability measures requires a regularity condition stronger than convergence of empirical measures in the Wasserstein topology. In particular, following the smooth-density construction of $\mathcal{P}^\infty(M)$, we consider

$$\mathcal{P}^\infty(M) := \left\{ q d\text{vol}_M : q \in C^\infty(M), q > 0, \int_M q d\text{vol}_M = 1 \right\}. \quad (153)$$

Thus, in order for an architecture-generated probability measure to belong to $\mathcal{P}^\infty(M)$, it is necessary to establish that its pushforward measure admits a smooth and strictly positive density with respect to $d\text{vol}_M$. A simple pathological example that illustrates this necessity may be this: $f_\theta(\epsilon) = 0 \forall \epsilon \in X$, that is, a neural network that outputs 0 for every input from the latent space. Needless to say, the probability distribution generated by this network will be $q_\theta = \delta_0$, for any parameter θ . This distribution is obviously atomic and thus is certainly not smooth. So, we must discuss the conditions under which architecture-generated distributions are smooth and impose these conditions as assumptions on which the manifold construction presented in this paper necessarily relies.

Let the prior distribution be absolutely continuous with respect to the volume measure on $\mathcal{Z}$, with density p_{prior} , (for example, it could be the Gaussian distribution) and define

$$\mu_\theta := (f_\theta)_\# p_{\text{prior}}. \quad (154)$$

The empirical measures

$$\mu_\theta^n = \frac{1}{n} \sum_{i=1}^n \delta_{f_\theta(\epsilon_i)}, \quad \epsilon_i \stackrel{\text{i.i.d.}}{\sim} p_{\text{prior}}, \quad (155)$$

may converge to μ_θ as $n \rightarrow \infty$ in an appropriate weak or Wasserstein sense. Such convergence, however, does not itself imply that μ_θ possesses a smooth density. Indeed, each finite empirical measure μ_θ^n is atomic, and Wasserstein convergence alone does not preserve smoothness or absolute continuity. Consequently, membership of μ_θ in $\mathcal{M}^{p_{\text{prior}}}$ must be established separately through regularity assumptions on the prior and the generator.

Dimensional and rank conditions. A necessary dimensional condition for f_θ to generate a distribution that is locally absolutely continuous with respect to the d -dimensional volume measure is

$$m \geq d. \quad (156)$$

If $m < d$, then the differential

$$D_\epsilon f_\theta(\epsilon) : \mathbb{R}^m \longrightarrow \mathbb{R}^d \quad (157)$$

cannot have rank d . Generically, the image of the generator is then lower-dimensional in M , and the resulting pushforward measure is singular with respect to $d\text{vol}_M$.

We therefore impose the full-rank condition

$$\text{rank } D_{\epsilon} f_{\theta}(\epsilon) = d \quad (158)$$

for all relevant ϵ and θ ²². Equivalently, f_{θ} is required to be a smooth submersion. In Euclidean coordinates this condition may be written as

$$D_{\epsilon} f_{\theta}(\epsilon) D_{\epsilon} f_{\theta}(\epsilon)^{\top} \succ 0, \quad (159)$$

or, equivalently,

$$\sigma_{\min}(D_{\epsilon} f_{\theta}(\epsilon)) > 0. \quad (160)$$

The distinction between $D_{\epsilon} f_{\theta}$ and the parameter Jacobian $D_{\theta} f_{\theta}$ is important. The former controls the local regularity and dimensional nondegeneracy of the generated probability measure, whereas the latter determines architecture-realizable parameter perturbations and enters the definition of the neural tangent kernel.

Equal-dimensional case The regularity statement is particularly transparent when $m = d$ and f_{θ} is a smooth diffeomorphism. Suppose

$$p_{\text{prior}} \in C^{\infty}(\mathbb{R}^m), \quad p_{\text{prior}} > 0, \quad (161)$$

and suppose

$$f_{\theta} : C^{\infty}(\mathbb{R}^m) \longrightarrow C^{\infty}(\mathbb{R}^d) \quad (162)$$

is a smooth diffeomorphism. Then the change-of-variables formula gives, in local coordinates,

$$q_{\theta}(y) = p_{\text{prior}}(f_{\theta}^{-1}(y)) |\det D_{\epsilon} f_{\theta}(f_{\theta}^{-1}(y))|^{-1}. \quad (163)$$

Since f_{θ}^{-1} is smooth and

$$\det D_{\epsilon} f_{\theta}(\epsilon) \neq 0, \quad (164)$$

Eq. (163) implies

$$q_{\theta} \in C^{\infty}(\mathbb{R}^d). \quad (165)$$

Moreover, strict positivity of p_{prior} implies

$$q_{\theta}(y) > 0 \quad (166)$$

for every $y \in M$. Hence

$$(f_{\theta})_{\#} p_{\text{prior}} = q_{\theta} d\text{vol}_{\mathbb{R}^d} \in \mathcal{P}^{\infty}(\mathbb{R}^d). \quad (167)$$

Higher-dimensional latent spaces. When $m > d$, injectivity of f_{θ} is neither expected nor required. Under the full-rank condition

$$\text{rank } D_{\epsilon} f_{\theta} = d, \quad (168)$$

the level set

$$f_{\theta}^{-1}(y) \quad (169)$$

is, locally, an $(m - d)$ -dimensional submanifold. The coarea formula yields a pushforward density of the schematic form

$$q_{\theta}(y) = \int_{f_{\theta}^{-1}(y)} \frac{p_{\text{prior}}(\epsilon)}{J_{f_{\theta}}(\epsilon)} d\mathcal{H}^{m-d}(\epsilon), \quad (170)$$

where

$$J_{f_{\theta}}(\epsilon) := \sqrt{\det(D_{\epsilon} f_{\theta}(\epsilon) D_{\epsilon} f_{\theta}(\epsilon)^{\top})} \quad (171)$$

is the normal Jacobian and $\mathcal{H}^{m-d}$ denotes the $(m - d)$ -dimensional Hausdorff measure on the fiber.

The submersion condition ensures that

$$J_{f_{\theta}}(\epsilon) > 0. \quad (172)$$

To conclude that the fiber integral in Eq. (170) defines a smooth function of y , additional global regularity conditions are required, such as suitable properness, compactness of fibers, or decay and integrability conditions permitting differentiation under the fiber integral. Under such assumptions, together with smoothness of p_{prior} and f_{θ} , one obtains

$$q_{\theta} \in \mathcal{P}^{\infty}(\mathbb{R}^d). \quad (173)$$

²²A question that follows from this is under what configurations of the neural network this condition holds. A set of sufficient conditions would be that the hidden dimension of every hidden layer must be at least d and that every weight matrix must be full rank.

Strict positivity. Smoothness of q_θ does not by itself imply strict positivity. To obtain

$$q_\theta(y) > 0 \quad \text{for every } y \in \mathbb{R}^d, \quad (174)$$

the generator must have sufficient global coverage of $\mathbb{R}^d$, the output space. A natural sufficient condition is surjectivity,

$$f_\theta(\mathbb{R}^m) = \mathbb{R}^d, \quad (175)$$

together with

$$p_{\text{prior}}(\epsilon) > 0 \quad (176)$$

on the relevant latent domain and the preceding nondegeneracy assumptions. Under these conditions, every $y \in M$ possesses a nonempty regular fiber, and the coarea representation yields a positive density.

Accordingly, a sufficient regularity regime for the architecture-generated measures may be summarized by the assumptions

$$m \geq d, \quad (177)$$

$$p_{\text{prior}} \in C^\infty(\mathbb{R}^m), \quad p_{\text{prior}} > 0, \quad (178)$$

$$f_\theta \in C^\infty(\mathbb{R}^m, \mathbb{R}^d), \quad (179)$$

$$\text{rank } D_\epsilon f_\theta(\epsilon) = d, \quad (180)$$

$$f_\theta(\mathbb{R}^m) = \mathbb{R}^d, \quad (181)$$

together with appropriate properness, compact-fiber, or decay and integrability conditions when required by the underlying spaces. Under these assumptions,

$$(f_\theta)_\# p_{\text{prior}} = q_\theta d\text{vol}_{\mathbb{R}^d}, \quad q_\theta \in \mathcal{P}^\infty(\mathbb{R}^d), \quad q_\theta > 0, \quad (182)$$

and consequently

$$(f_\theta)_\# p_{\text{prior}} \in \mathcal{P}^\infty(\mathbb{R}^d). \quad (183)$$

This regularity requirement is stronger than the assumptions necessary for the drifting algorithm itself. Its role is specifically geometric: it restricts attention to the nondegenerate architecture-generated distributions that lie in the smooth infinite-dimensional manifold $\mathcal{M}_f^{p_{\text{prior}}}$, thereby admitting an interpretation within a genuine smooth-manifold framework.

9.5 Proof of Fréchet Differentiability of the Mapping $\hat{\Phi} : \mathbb{R}^p \longrightarrow H_{0,q_0}^s$

With the differential of Φ defined in section 4.1.3, it is incumbent upon us to prove Fréchet differentiability. The differential we defined in section 4.1.3 can only be considered a **candidate differential**, since it reveals what the differential, if it exists, must be. But since we have not proven Fréchet differentiability, we cannot claim that that weak, distributional differentiation is a genuine differential. To promote the weak calculation to a genuine Fréchet derivative, it remains to verify Eq. (28) under the stated regularity assumptions. Equivalently, sufficient joint regularity of $(\theta, x) \mapsto q_\theta(x)$ together with appropriate uniform Sobolev bounds ensures that the remainder after linearization is $o(\|h\|_{\mathbb{R}^p})$ in the chosen H^s norm. Fréchet differentiability is likewise a necessary condition for the the image of the Constant Rank Theorem invoked later to give a genuine smooth immersed submanifold. Thus, this theorem is also indispensable for our manifold construction of probability distributions and must be established before we invoke the Constant Rank Theorem in section 4.1.4.

Let us restate the definition of Fréchet differentiability on the appropriate mapping $\hat{\Phi}$ that we wish to prove:

Theorem 28 (The composition mapping $\hat{\Phi} = \chi \circ \Phi$ from the parameter space $\mathbb{R}^p$ to the Hilbert manifold H_{0,q_0}^s is Fréchet differentiable).

$$\lim_{\|h\| \rightarrow 0} \frac{\left\| \hat{\Phi}(\theta + h) - \hat{\Phi}(\theta) - D\hat{\Phi}_\theta[h] \right\|_{H^s}}{\|h\|_{\mathbb{R}^p}} = 0. \quad (184)$$

Proof. Before we prove this limit, it would be helpful to expand out two things: the definition of the Hilbert space norm and the candidate differential of the $\widehat{\Phi}$ mapping. While there is an alternative Fourier definition for the Hilbert norm, we adopt

$$\|f\|_{H^s} := \left(\sum_{|\alpha| \leq s} \|D^\alpha f\|_{L^2}^2 \right)^{\frac{1}{2}},$$

where α is multindex consisting of a tuple of integers $\alpha = (\alpha_1, \alpha_2, \dots, \alpha_d)$, with $|\alpha| = \alpha_1 + \alpha_2 + \dots + \alpha_d$, which denotes a multivariate partial derivative with the i^{th} variable α_i times..

So, for the limit to be true, it suffices to prove that the numerator is of order $O(\|h\|_{\mathbb{R}^p}^a)$, with $a > 1$, so that as the Euclidean norm of h approaches 0, the numerator decays faster, leading the limit to converge to 0. Or equivalently, it is of order $o(\|h\|_{\mathbb{R}^p})$ as it decays.

Let us also discuss the current candidate differential of $\widehat{\Phi}$. Recall that $\widehat{\Phi} = \chi_{q_0} \circ \Phi$, where

$$\widehat{\Phi}(\theta) = \log \left(\frac{q_\theta}{q_0} \right) - \int_{\mathbb{R}^d} \log \left(\frac{q_\theta(x)}{q_0(x)} \right) q_0(x) dx.$$

Differentiating in the direction $h \in \mathbb{R}^p$ and applying the chain rule gives

$$D\widehat{\Phi}(\theta)[h] = \frac{d\Phi_\theta[h]}{q_\theta} - \int_{\mathbb{R}^d} \frac{d\Phi_\theta[h](x)}{q_\theta(x)} q_0(x) dx.$$

Using the previously established identity

$$d\Phi_\theta[h] = -\nabla \cdot (q_\theta J_\theta f_\theta h),$$

we obtain

$$D\widehat{\Phi}(\theta)[h] = -\frac{\nabla \cdot (q_\theta J_\theta f_\theta h)}{q_\theta} + \int_{\mathbb{R}^d} \frac{\nabla \cdot (q_\theta J_\theta f_\theta h)}{q_\theta} q_0 dx.$$

Returning to the Hilbert norm. Note that, for $|\alpha| \geq 1$, all the centering ratios, which are constants, will vanish under differentiation. So, it is helpful to break down the H^s norm of the numerator into two parts, and consider them separately:

$$\left\| \widehat{\Phi}(\theta + h) - \widehat{\Phi}(\theta) - D\widehat{\Phi}_\theta[h] \right\|_{H^s}^2 = \left(\left\| \widehat{\Phi}(\theta + h) - \widehat{\Phi}(\theta) - D\widehat{\Phi}_\theta[h] \right\|_{L^2}^2 + \sum_{1 \leq |\alpha| \leq s} \|D^\alpha [\widehat{\Phi}(\theta + h) - \widehat{\Phi}(\theta) - D\widehat{\Phi}_\theta[h]]\|_{L^2}^2 \right)$$

Case 1: $1 \leq |\alpha| \leq s$

Let us first consider the latter and try to bound it.

$$\sum_{1 \leq |\alpha| \leq s} \|D^\alpha [\widehat{\Phi}(\theta + h) - \widehat{\Phi}(\theta) - D\widehat{\Phi}_\theta[h]]\|_{L^2}^2$$

Since all log-ratio term vanishes in differentiation, notice that this is simply

$$\sum_{1 \leq |\alpha| \leq s} \|D^\alpha \left[\log \frac{q_{\theta+h}}{q_0} - \log \frac{q_\theta}{q_0} + \frac{\nabla \cdot (q_\theta J_\theta f_\theta h)}{q_\theta} \right]\|_{L^2}^2,$$

which cancels to

$$\sum_{1 \leq |\alpha| \leq s} \|D^\alpha \left[\log q_{\theta+h} - \log q_\theta + \frac{\nabla \cdot (q_\theta J_\theta f_\theta h)}{q_\theta} \right]\|_{L^2}^2,$$

Paying attention to the first two terms, we can rearrange it as

$$\log q_{\theta+h} - \log q_\theta = \log \left(1 + \frac{q_{\theta+h} - q_\theta}{q_\theta} \right)$$

Let us define

$$z_h := \frac{q_{\theta+h} - q_\theta}{q_\theta},$$

so that

$$\log \left(\frac{q_{\theta+h}}{q_\theta} \right) = \log(1 + z_h).$$

Recall that the intrinsic directional differential obtained earlier is

$$d\Phi_\theta[h] = -\nabla \cdot (q_\theta J_\theta f_\theta h).$$

Hence, define the corresponding linear term

$$\ell_h := \frac{d\Phi_\theta[h]}{q_\theta} = -\frac{\nabla \cdot (q_\theta J_\theta f_\theta h)}{q_\theta}.$$

The relevant remainder may therefore be decomposed as

$$\log(1 + z_h) - \ell_h = [\log(1 + z_h) - z_h] + [z_h - \ell_h].$$

We first control the nonlinear logarithmic remainder. Recall the Taylor expansion

$$\log(1 + z) = z - \frac{z^2}{2} + \frac{z^3}{3} - \dots, \quad |z| < 1.$$

Thus,

$$\log(1 + z) - z = O(z^2) \quad \text{as } z \rightarrow 0.$$

Suppose that, for sufficiently small h ,

$$\|z_h\|_{H^s} = O(\|h\|_{\mathbb{R}^p}).$$

Since $s > d/2$, the Sobolev space $H^s(\mathbb{R}^d)$ is a Banach algebra. In particular, there exists a constant $C > 0$ such that

$$\|fg\|_{H^s} \leq C\|f\|_{H^s}\|g\|_{H^s}.$$

Consequently, the quadratic and higher-order terms in the Taylor expansion satisfy

$$\|\log(1 + z_h) - z_h\|_{H^s} \leq C\|z_h\|_{H^s}^2 = O(\|h\|_{\mathbb{R}^p}^2).$$

Equivalently, using the derivative characterization of the Sobolev norm,

$$\boxed{\sum_{|\alpha| \leq s} \|D^\alpha [\log(1 + z_h) - z_h]\|_{L^2} = O(\|h\|_{\mathbb{R}^p}^2)}.$$

Now this is exactly what we want. We have now shown that this part of the Sobolev norm is of order higher order decay.

Case 2: $|\alpha| = 0$

Let us consider the first term, the term when $|\alpha| = 0$, i.e., when no derivative is yet taken. Since this is just one function, its L2 norm is easy to compute.

For the zeroth-order contribution, namely $|\alpha| = 0$, the Sobolev term is simply the L^2 norm of the Fréchet remainder itself. Expanding the definition of $\widehat{\Phi}$ gives

$$\begin{aligned} R_h^{(0)} &= \log\left(\frac{q_{\theta+h}}{q_0}\right) - \int_{\mathbb{R}^d} \log\left(\frac{q_{\theta+h}(x)}{q_0(x)}\right) q_0(x) dx \\ &\quad - \log\left(\frac{q_\theta}{q_0}\right) + \int_{\mathbb{R}^d} \log\left(\frac{q_\theta(x)}{q_0(x)}\right) q_0(x) dx \\ &\quad - \frac{d\Phi_\theta[h]}{q_\theta} + \int_{\mathbb{R}^d} \frac{d\Phi_\theta[h](x)}{q_\theta(x)} q_0(x) dx. \end{aligned}$$

Define

$$A_h := \log\left(\frac{q_{\theta+h}}{q_\theta}\right) - \frac{d\Phi_\theta[h]}{q_\theta}.$$

Then the preceding expression may be written compactly as

$$R_h^{(0)} = A_h - \int_{\mathbb{R}^d} A_h(x) q_0(x) dx = A_h - \mathbb{E}_{q_0}[A_h].$$

Now define

$$z_h := \frac{q_{\theta+h} - q_\theta}{q_\theta},$$

so that

$$\log\left(\frac{q_{\theta+h}}{q_\theta}\right) = \log(1 + z_h).$$

Hence

$$A_h = \log(1 + z_h) - \frac{d\Phi_\theta[h]}{q_\theta}.$$

Decomposing the latter expression,

$$A_h = [\log(1 + z_h) - z_h] + \left[z_h - \frac{d\Phi_\theta[h]}{q_\theta} \right].$$

From the Taylor expansion of $\log(1 + z)$ around $z = 0$,

$$\log(1 + z) - z = O(z^2).$$

Thus, provided

$$\|z_h\|_{H^s} = O(\|h\|_{\mathbb{R}^p}),$$

the Sobolev algebra property for $s > d/2$ implies

$$\|\log(1 + z_h) - z_h\|_{H^s} = O(\|h\|_{\mathbb{R}^p}^2).$$

Together with the linearization remainder estimate

$$\left\| z_h - \frac{d\Phi_\theta[h]}{q_\theta} \right\|_{H^s} = o(\|h\|_{\mathbb{R}^p}),$$

we obtain

$$\|A_h\|_{L^2} = o(\|h\|_{\mathbb{R}^p}).$$

Finally, since the centering map

$$C_{q_0}(f) := f - \mathbb{E}_{q_0}[f]$$

is linear and bounded on the relevant function class, it preserves the asymptotic order. Therefore,

$$\|R_h^{(0)}\|_{L^2} = \|A_h - \mathbb{E}_{q_0}[A_h]\|_{L^2} = o(\|h\|_{\mathbb{R}^p}).$$

Hence the $|\alpha| = 0$ contribution satisfies the required Fréchet remainder estimate.

Thus, we have now proven that both parts in the Hilbert norm, $|\alpha| = 0$ and $1 \leq |\alpha| \leq s$ are of $o(\|h\|_{\mathbb{R}^p})$ order as $\|h\|_{\mathbb{R}^p}$ decays to 0. Thus, the limit at hand converges to 0. Fréchet differentiability is proven. $\square$

9.6 The regular parameter set is an open subset

Lemma 29 (Injectivity via the Gram Matrix). *Let H be a real Hilbert space and let $T : \mathbb{R}^p \rightarrow H$ be linear. Let $e_1, \dots, e_p$ denote the standard basis of $\mathbb{R}^p$ and define the Gram matrix $G \in \text{Sym}(p, \mathbb{R})$ by $G_{ij} := \langle Te_i, Te_j \rangle_H$. Then*

$$\ker T = \{0\} \iff \det G \neq 0.$$

Proof. For $v = \sum_i v_i e_i \in \mathbb{R}^p$,

$$v^\top G v = \sum_{i,j} v_i v_j \langle Te_i, Te_j \rangle_H = \left\langle \sum_i v_i Te_i, \sum_j v_j Te_j \right\rangle_H = \|Tv\|_H^2 \geq 0,$$

so G is symmetric positive semi-definite, and $Tv = 0 \iff v^\top G v = 0 \iff Gv = 0$ (the last equivalence holding since $G \succeq 0$). Hence $\ker T = \ker G$ as subspaces of $\mathbb{R}^p$. A symmetric matrix has trivial kernel if and only if it is nonsingular, i.e. $\det G \neq 0$. The claim follows. $\square$

Lemma 30 (Continuity of the Gram Map). *Suppose $\theta \mapsto D\hat{\Phi}_\theta \in \mathcal{L}(\mathbb{R}^p, H_{0,q_0}^s(\mathbb{R}^d))$ is continuous in operator norm on $\mathbb{R}^p$ (this is precisely the continuity hypothesis used to establish Fréchet differentiability of $\hat{\Phi}$ in Theorem 21 / Appendix 9.5). Define $G : \mathbb{R}^p \rightarrow \text{Sym}(p, \mathbb{R})$ by*

$$G(\theta)_{ij} := \langle D\hat{\Phi}_\theta[e_i], D\hat{\Phi}_\theta[e_j] \rangle_{H^s}, \quad i, j = 1, \dots, p.$$

Then $\theta \mapsto G(\theta)$ is continuous.

Proof. Fix $i, j \in \{1, \dots, p\}$ and $\theta_0 \in \mathbb{R}^p$. Write $A_\theta := D\hat{\Phi}_\theta[e_i]$ and $B_\theta := D\hat{\Phi}_\theta[e_j]$, elements of $H_{0,q_0}^s(\mathbb{R}^d)$. By bilinearity of the inner product,

$$G(\theta)_{ij} - G(\theta_0)_{ij} = \langle A_\theta - A_{\theta_0}, B_\theta \rangle_{H^s} + \langle A_{\theta_0}, B_\theta - B_{\theta_0} \rangle_{H^s}.$$

By the Cauchy–Schwarz inequality in $H_{0,q_0}^s(\mathbb{R}^d)$,

$$|G(\theta)_{ij} - G(\theta_0)_{ij}| \leq \|A_\theta - A_{\theta_0}\|_{H^s} \|B_\theta\|_{H^s} + \|A_{\theta_0}\|_{H^s} \|B_\theta - B_{\theta_0}\|_{H^s}.$$

By the operator-norm continuity hypothesis, $\|A_\theta - A_{\theta_0}\|_{H^s} \rightarrow 0$ and $\|B_\theta - B_{\theta_0}\|_{H^s} \rightarrow 0$ as $\theta \rightarrow \theta_0$; in particular $\theta \mapsto \|B_\theta\|_{H^s}$ is continuous at θ_0 , hence bounded on a neighborhood of θ_0 . Both terms on the right therefore vanish as $\theta \rightarrow \theta_0$, so $G(\theta)_{ij} \rightarrow G(\theta_0)_{ij}$. Since i, j, θ_0 were arbitrary, G is continuous on $\mathbb{R}^p$. $\square$

Theorem 31 (Openness of the Regular Parameter Set). *The regular parameter set*

$$\mathbb{R}_{reg}^p := \{ \theta \in \mathbb{R}^p : d\Phi_\theta \text{ is injective} \}$$

is an open subset in $\mathbb{R}^p$.

Proof. Since $\chi_{q_0} : \mathcal{P}_{q_0}^s(\mathbb{R}^d) \rightarrow H_{0,q_0}^s(\mathbb{R}^d)$ is a diffeomorphism (Section 4.1.2), $d\Phi_\theta$ is injective if and only if $D\hat{\Phi}_\theta = D(\chi_{q_0} \circ \Phi)_\theta$ is injective. Applying Lemma 29 with $T = D\hat{\Phi}_\theta$ and $H = H_{0,q_0}^s(\mathbb{R}^d)$,

$$\theta \in \mathbb{R}_{reg}^p \iff \det G(\theta) \neq 0,$$

where G is the Gram map of Lemma 30. Thus

$$\mathbb{R}_{reg}^p = (\det \circ G)^{-1}(\mathbb{R} \setminus \{0\}).$$

The determinant $\det : \text{Sym}(p, \mathbb{R}) \rightarrow \mathbb{R}$ is a polynomial in the matrix entries, hence continuous; G is continuous by Lemma 30; so $\det \circ G$ is continuous as a composition of continuous maps. Since $\mathbb{R} \setminus \{0\}$ is open in $\mathbb{R}$, its preimage $\mathbb{R}_{reg}^p$ under the continuous map $\det \circ G$ is open in $\mathbb{R}^p$. $\square$

Remark 32. *The continuity hypothesis in Lemma 30 is exactly the estimate needed to upgrade the Gateaux derivative $d\Phi_\theta$ of Section 4.1.3 to a genuine Fréchet derivative in Theorem 21; no additional regularity beyond that already required for Fréchet differentiability is needed to obtain openness of $\mathbb{R}_{reg}^p$. By contrast, density of $\mathbb{R}_{reg}^p$ in $\mathbb{R}^p$ requires the strictly stronger hypothesis that $\theta \mapsto G(\theta)$ be real-analytic, together with the identity theorem for real-analytic functions, and is addressed separately as in future work section 8.1.*